



About calibration of car-following dynamics of automated and human-driven vehicles: Methodology, guidelines and codes

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ABSTRACT

A comprehensive literature review reveals that there exist lots of ambiguities, confusion and even contradictions in setting a car-following calibration problem. In particular, confusion arises in the selection of measure of performances and goodness-of-fit functions. In this study, a methodology to compare and rank objective functions is thus proposed, which is based on Pareto-efficiency and on indifference curves. The methodology has been applied to all objective functions used in the field literature so far (and to new ones), in a vast set of calibration experiments. The experiments involved two car-following models and two adaptive cruise control (ACC) algorithms, and four different datasets, including both automated and human-driven vehicles trajectories. Since results are consistent among all the calibration experiments, a sound and robust guideline to calibrate car-following dynamics has been proposed. It includes recommendation about what calibration settings should be avoided and what are to be adopted. On the one hand, a general agreement on a sound calibration setting for car-following models is deemed necessary for comparing results from different studies which use different models and datasets. On the other hand, any new car-following model or objective function being developed in the future shall be compared with existing ones in a fair and impartial manner. For these reasons, and to promote and enable transparent and reproducible research, codes and data from this study are shared with the community.

1. Introduction

Models are just approximations of reality, and car-following (CF) models are no exception. Thus, we will never have a CF model delivering a zero error, whatever *goodness-of-fit function* (GoF) and *measure of performance* (MoP), or combination of MoPs, are used. This simple fact has profound implications on the calibration of CF dynamics.

As models are imperfect – and only because models are imperfect – it is not the same if we calibrate them on a different MoP and/or a different GoF. Consequentially, CF model calibration is prone to ambiguities, confusions, and even contradictions, as unfortunately confirmed by our state-of-the-art review, despite the fact that CF modelling is unarguably one of the oldest and most frequently studied topics in traffic flow theory. These problems lead to a subjectivity of CF modelling research and impedes its fast progress. Particularly, the wide spectrum of calibration settings in the literature makes it impossible to quantitatively compare and sift results from different studies using different models and datasets (e.g., exposing the bad while retaining the good), and hence undermines the transparency

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and reproducibility of research.

This paper aims at changing this trend by providing a sound recommendation on calibration settings based on a vast experimental analysis and on previous theoretical results.

Furthermore, the results of this study are reproducible. Methodology, codes and data underlying the study and its recommendations, are shared with the scientific community. Thus, future developments – being them a new or different *GoF*, a new model or a new trajectory dataset – could be simply plugged-in in the study framework and compared in a fair and impartial manner.

To inform the experimental analysis in this study, a comprehensive review of how *CF* dynamics have been calibrated in the recent literature has been carried out. A state of confusion, contradictions, and even malpractices emerged (Sections 2 and 3). We have then tested and ranked all the combinations of *MoPs* and *GoFs* used in the literature so far (Section 4). To this aim four different well-known car-following models/adaptive cruise control algorithms (ACC) have been calibrated on four trajectory datasets, including car-following dynamics of both human-driven vehicles and automated vehicles (i.e. vehicles equipped with ACC).

Grounded on this experimental analysis and on previous theoretical results (see Punzo and Montanino, 2016), two types of recommendation are provided in the study (Section 5).

First, a number of calibration settings, which are inappropriate but often used in the field literature, have been identified. In the epistemological framework we assume in this study, this result is deemed to be definitive. One sound refutation is sufficient to dismiss a general theory, indeed (see the essay by Popper, 1963, on conjectures and refutations). All the more so, if a calibration setting appears inappropriate in the vast majority of experimental scenarios (i.e. with different car-following models and datasets), it cannot have general validity. For instance, hereafter we will report that $RMSE(v)$ is worse than $RMSE(s)$ in 93.5% of cases. Therefore, in a future experiment, one should avoid using a setting that has repeatedly failed in the literature. Although there may exist some probability that such a setting worked in a specific experimental scenario, it would remain to be demonstrated that it was the most efficient in that scenario (see the next paragraph).

Then a sound recommendation about what *MoP* and *GoF* should be used for *CF* dynamics calibration is given. Unlike the first recommendation, we are aware that this one is not definitive, since experimental results, though wide, will never be a proof. There may be other experiments carried out on different models or data, or simply proposing new calibration settings that may provide different or better results than those in this study. However, these new results would at most impact the optimality, but not the soundness of the recommendation in this study. Furthermore, the superiority of any new result can only be established by comparing it with previous efficient settings (like the recommended settings in this study) in an impartial and reproducible manner e.g., through the methodological framework (and the codes) provided here.

Although the main focus of the paper is on calibration settings, the experimental framework also allowed interesting comparisons of models and trajectory datasets of automated and human-driven vehicles to be made (see Section 5).

What's more, this paper is reproducible, and anyone who is interested in checking any part of it can generate the same content in the open source RStudio (RStudio, 2020).

2. Confusion, contradictions and malpractices: the state of the art of *CF* model calibration

A general formulation of the *CF* model calibration problem is as follows (Sharma et al., 2019a):

$$\begin{aligned} MoP^{sim} &= F(\beta) \\ \text{minimize } &f(MoP^{obs}, MoP^{sim}) \\ \text{subject to : } &LB_{\beta} \leq \beta \leq UB_{\beta}, G(\beta) \leq 0 \end{aligned} \quad (1)$$

where β is a vector of *CF* model parameters (number of parameters can change depending on the *CF* model adopted) to be calibrated; $F(\cdot)$ is a car-following model, that is a function of β ; LB_{β} and UB_{β} represent the lower and the upper bound for the parameters in β , respectively; $G(\cdot)$ is a vector of constraint functions; MoP^{obs} and MoP^{sim} represent observed and simulated *MoP*, respectively; $f(\cdot)$ is a *GoF* function; and $f(MoP^{obs}, MoP^{sim})$ is the optimization objective function. The aim of *CF* model calibration is to minimize $f(MoP^{obs}, MoP^{sim})$. In this general formulation, there are six important elements related to model calibration: the model itself, data and data/model integration scheme, the *MoP*, the *GoF*, and the optimization algorithm.

Regarding the data and data/model integration scheme, a consensus has been reached in the field literature. First, error and bias in the data need to be checked (Punzo et al. 2011; Treiber and Kesting 2013). More specifically, when pre-processing data, care must be taken to avoid error amplification in the differentiation process (Punzo et al., 2011), and accumulation in the integration process (Punzo et al., 2011, Punzo and Montanino, 2016),¹ e.g., several studies have pointed out errors in the NGSIM datasets (Duret et al., 2008; Thiemann et al., 2008; Punzo et al., 2011) and proposed methodologies to filter them out (Coifman and Li, 2017; Montanino and Punzo, 2013, 2015).

Remark 1. If several measures are independently collected through separate devices (e.g. position from GPS, relative velocity from radar, acceleration from an inertial platform), these measurements need to be “internally consistent”. Specifically, spacing (speed) measurements need to be equal to the integral of speed (acceleration) measurements (see the formal definition of internal consistency

¹ Since differentiation amplifies high frequency signal components, noise is usually amplified in the differentiation process from spacing to speed to acceleration. Conversely, non-white noise in acceleration (speed) accumulates in speed (spacing) in the integration process.

Table 1State-of-the-art about objective functions in car-following model calibration (see [Appendix B](#) for some notes about papers classification).

GoF(MoP)	Literature Review		
	2005–2011	2012–2015	2016–2020
$RMSE(s)$	Punzo and Simonelli (2005)	Punzo et al. (2012)	Punzo and Montanino (2016) , Rhoades et al. (2016) , Yu et al. (2016) , Zhou et al. (2017) , Hammit et al. (2018) , Wang et al. (2018) , Yu et al. (2018) , Park et al. (2020) , Xu and Laval (2020) , Montanino and Punzo (2021) , Montanino et al. (2021) .
$RMSE(v)$	Punzo and Simonelli (2005) , Hoogendoorn and Hoogendoorn (2010a, 2010b) , Chong et al. (2011) , Hoogendoorn et al. (2010)	Punzo et al. (2012) , Rahman et al. (2015) , Montanino et al. (2012) , Higgs and Abbas (2015) , van Hinsbergen et al. (2015) , Vieira da Rocha et al. (2015)	Gunter et al. (2020) , Huang et al. (2018a) , Li et al. (2016, 2020) , Ngoduy et al. (2019) , Punzo and Montanino (2016) , Ye et al. (2019) , Zhao and Zhang (2020)
$RMSE(a)$		Kim et al. (2013)	Hao et al. (2016) , Pei et al. (2016) , Li et al. (2018) , James et al. (2019) , Xu et al. (2019)
$RMSE(\sigma_v)$			Huang et al. (2018b) , Tian et al. (2019)
$RMSPE(s)$	Kesting and Treiber (2008) , Wang et al. (2011)	Yu and Shi (2014) , Saifuzzaman et al. (2015)	Saifuzzaman et al. (2017) , Wang et al. (2017) , Zhu et al. (2018b) , Sharma et al. (2019a, 2019b) , Kurtc and Treiber (2020)
$RMSPE(v)$			Papathanasopoulou et al. (2016)
$RMSPE(a)$		Chong et al. (2013)	
$RMSPE(\sigma_v)$			Tian et al. (2016)
$U(s)$	Ossen and Hoogendoorn (2008)	Punzo et al. (2012) , Jin et al. (2014)	Cattin et al. (2019)
$U(v)$	Ossen and Hoogendoorn (2008)	Punzo et al. (2012) , Jin et al. (2014)	Yang et al. (2016) , Cattin et al. (2019)
$U(a)$		Jin et al. (2014)	
$U(s, v)$	Ossen and Hoogendoorn (2009) with $\alpha = \beta$, Kim, and Mahmassani (2011) with $\alpha = \beta$	Sangster et al. (2013) with $\alpha = \beta$	Zhong et al. (2016) with $\alpha = \beta$, Mamouei et al. (2016) with $\alpha = \beta$, Wang et al. (2019) with $\alpha = \beta$, Zhang et al. (2019) with $\alpha = 1, \beta = 0.5$, Chen et al. (2020) with $\alpha = 1, \beta = 0.5$
$MAE(s)$		Punzo et al. (2012) , Jin et al. (2014)	He et al. (2016)
$MAE(v)$		Punzo et al. (2012) , Milanes and Shladover (2014) , Jin et al. (2014)	
$MAE(a)$		Jin et al. (2014)	Peng et al. (2020)
$MAPE(s)$			Zheng et al. (2016)
$MAPE(a)$		Chong et al. (2013)	
$\widetilde{RMSPE}(s)$	Kesting and Treiber (2008)	Monteil et al. (2014)	Kurtc and Treiber (2020) , Hamdar et al. (2020)
$\widetilde{RMSPE}(v)$		Hamdar et al. (2015)	Schorr et al. (2017)
$NRMSE(s)$	Kesting and Treiber (2008)		Zhu et al. (2018a, 2018b, 2020) , Kurtc and Treiber (2020) ,
$NRMSE(v)$		Papathanasopoulou and Antoniou (2015)	

in [Punzo et al., 2011](#), later reported in [Treiber and Kesting, 2013](#)). In case they are not internally consistent, an appropriate methodology to filter data needs to be employed (see [Punzo et al., 2005](#); [Montanino and Punzo, 2015](#)).

Remark 2. When either one or more measurements are available (e.g. only spacing from GPS, or spacing from GPS and relative speed from radar), the integration/differentiation scheme of data needs to be the same adopted in model simulation. Otherwise, the residuals given by $f(MoP^{obs}, MoP^{sim})$ would be caused by the CF model approximation, as well as by a different integration/differentiation scheme between model and data (for integration schemes in CF models, see [Treiber and Kesting, 2013](#); and [Treiber and Kanagaraj, 2015](#). For a thorough introduction to errors and uncertainty in traffic simulation, see [Punzo and Montanino, 2020](#)).

Regarding optimization algorithms, a systematic and unbiased assessment of different algorithms (and calibration settings) was performed in [Punzo et al. \(2012\)](#), in a laboratory experiment. Synthetic trajectories were generated through a CF model, by assuming a given set of model parameter values. Model calibrations were run with the objective of rediscovering these pre-determined parameter values. Only genetic algorithm (GA) and combined scatter-search and gradient-based NLP solver (LINDO OptQuest) were able to perform satisfactorily. Downhill simplex and other algorithms, which were not reported in that paper for the sake of brevity (i.e. Interior Point, Trust Region Reflective and Sequential Quadratic Programming solvers), failed to rediscover the predetermined parameter values. As claimed in the Introduction, we consider the proved inability of some algorithms to successfully calibrate a CF model – in a systematic and fair test environment – as a sufficient proof to dismiss these algorithms. Meanwhile, the good performances of GA and OptQuest suggest that choosing them as a general optimization tool for CF calibration is reasonable, though not necessarily optimal or valid in every circumstance, e.g., for CF calibration against macroscopic traffic data such as loop detector data or average travel times (e.g. [Brockfeld et al., 2003](#); [Ciuffo et al., 2008](#); [Cascetta et al., 2011](#)). For a review of optimization algorithms used in CF model calibration see [Rafati Fard and Shariat Mohaymany \(2019\)](#).

Table 2Formulas of possible combinations of the 7 *GoFs* and the 4 *MoPs* from Table 1.

<i>GoF(MoP)</i>
$RMSE(s) = \sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [s^{sim}(t) - s^{obs}(t)]^2}$
$RMSE(v) = \sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [v^{sim}(t) - v^{obs}(t)]^2}$
$RMSE(a) = \sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [a^{sim}(t) - a^{obs}(t)]^2}$
$RMSE(\sigma_v) = \sqrt{(\sigma_v^{sim} - \sigma_v^{obs})^2}$
$RMSPE(s) = \sqrt{\frac{1}{T} \cdot \sum_{t=1}^T \left[\frac{s^{sim}(t) - s^{obs}(t)}{s^{obs}(t)} \right]^2}$
$RMSPE(v) = \sqrt{\frac{1}{ D } \cdot \sum_{t \in D} \left[\frac{v^{sim}(t) - v^{obs}(t)}{v^{obs}(t)} \right]^2}, D = \{t \in [1, T] : v^{obs}(t) \geq v_{th}\}$
$RMSPE(a) = \sqrt{\frac{1}{ D } \cdot \sum_{t \in D} \left[\frac{a^{sim}(t) - a^{obs}(t)}{a^{obs}(t)} \right]^2}, D = \{t \in [1, T] : a^{obs}(t) \geq a_{th}\}$
$RMSPE(\sigma_v) = \sqrt{[(\sigma_v^{sim} - \sigma_v^{obs}) / \sigma_v^{obs}]^2}$
$RMSPE(s, v) = \alpha \cdot RMSPE(s) + \beta \cdot RMSPE(v)$
$RMSPE(s, v, a) = \alpha \cdot RMSPE(s) + \beta \cdot RMSPE(v) + \gamma \cdot RMSPE(a)$
$RMSPE(s, \sigma_v) = \alpha \cdot RMSPE(s) + \beta \cdot RMSPE(\sigma_v)$
$U(s) = \frac{RMSE(s)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [s^{sim}(t)]^2 + \frac{1}{T} \cdot \sum_{t=1}^T [s^{obs}(t)]^2}}$
$U(v) = \frac{RMSE(v)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [v^{sim}(t)]^2 + \frac{1}{T} \cdot \sum_{t=1}^T [v^{obs}(t)]^2}}$
$U(a) = \frac{RMSE(a)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [a^{sim}(t)]^2 + \frac{1}{T} \cdot \sum_{t=1}^T [a^{obs}(t)]^2}}$
$U(s, v) = \alpha \cdot U(s) + \beta \cdot U(v)$
$U(s, v, a) = \alpha \cdot U(s) + \beta \cdot U(v) + \gamma \cdot U(a)$
$MAE(s) = \frac{1}{T} \cdot \sum_{t=1}^T s^{sim}(t) - s^{obs}(t) $
$MAE(v) = \frac{1}{T} \cdot \sum_{t=1}^T v^{sim}(t) - v^{obs}(t) $
$MAE(a) = \frac{1}{T} \cdot \sum_{t=1}^T a^{sim}(t) - a^{obs}(t) $
$MAPE(s) = \frac{1}{T} \cdot \sum_{t=1}^T \frac{ s^{sim}(t) - s^{obs}(t) }{s^{obs}(t)}$
$MAPE(v) = \frac{1}{T} \cdot \sum_{t=1}^T \frac{ v^{sim}(t) - v^{obs}(t) }{v^{obs}(t)}, D = \{t \in [1, T] : v^{obs}(t) \geq v_{th}\}$
$MAPE(a) = \frac{1}{T} \cdot \sum_{t=1}^T \frac{ a^{sim}(t) - a^{obs}(t) }{ a^{obs}(t) }, D = \{t \in [1, T] : a^{obs}(t) \geq a_{th}\}$
$MAPE(s, v) = \alpha \cdot MAPE(s) + \beta \cdot MAPE(v)$
$MAPE(s, v, a) = \alpha \cdot MAPE(s) + \beta \cdot MAPE(v) + \gamma \cdot MAPE(a)$
$\widetilde{RMSPE}(s) = \sqrt{\frac{1}{\sum_{t=1}^T s^{obs}(t)} \cdot \sum_{t=1}^T \frac{[s^{sim}(t) - s^{obs}(t)]^2}{s^{obs}(t)}}$
$\widetilde{RMSPE}(v) = \sqrt{\frac{T}{ D \cdot \sum_{t=1}^T v^{obs}(t)} \cdot \sum_{t \in D} \left[\frac{v^{sim}(t) - v^{obs}(t)}{v^{obs}(t)} \right]^2}, D = \{t \in [1, T] : v^{obs}(t) \geq v_{th}\}$
$\widetilde{RMSPE}(s, v) = \alpha \cdot \widetilde{RMSPE}(s) + \beta \cdot \widetilde{RMSPE}(v)$
$NRMSE(s) = \frac{RMSE(s)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [s^{obs}(t)]^2}}$
$NRMSE(v) = \frac{RMSE(v)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [v^{obs}(t)]^2}}$
$NRMSE(a) = \frac{RMSE(a)}{\sqrt{\frac{1}{T} \cdot \sum_{t=1}^T [a^{obs}(t)]^2}}$
$NRMSE(s, v) = \alpha \cdot NRMSE(s) + \beta \cdot NRMSE(v)$
$NRMSE(s, v, a) = \alpha \cdot NRMSE(s) + \beta \cdot NRMSE(v) + \gamma \cdot NRMSE(a)$

Therefore, as efficient optimization algorithms and consistent trajectory data and models are available – which makes a correct and unbiased analysis possible – in the rest of the paper we only focus on the problem of choosing a suitable objective function for calibration, i.e., an appropriate combination of *GoF* and *MoPs*. Thus, in the next section, we systematically and comprehensively review and then comment on the state of the art of the CF model calibration literature concerning *MoPs* and *GoFs*.

2.1. Literature review

Through Scopus®, we have implemented a comprehensive literature search on the calibration of *CF* dynamics, using “Car-following” and/or “adaptive cruise control” in combination with “calibration” as the search keywords (please see [Appendix A](#) for the Scopus® queries). The list of the target journals is presented in [Appendix A](#).

We found that 7 *GoFs* (*RMSE*, *RMSPE*, *Theil's U*, *MAE*, *MAPE*, *RMSPE*, *NRMSE*) and 4 *MoPs* (i.e., s , v , a , σ_v) were used in the literature on *CF* modelling. In total, 21 combinations of *GoFs* and *MoPs*, i.e., objective functions, were adopted as summarized in [Table 1](#). In [Table 2](#), the mathematical formulations of 32 possible *GoFs* and *MoPs* combinations, including those in [Table 1](#), are reported.

Literature review reported in [Table 1](#) covers the period from 2005 to 2020. Papers are grouped in three different time periods, to bring the evolution of consensus about the calibration settings to light, through the whole period. Specifically, the extension of periods was chosen in order to show the acceptance in the field literature of some relevant papers on the topic.

Remark 3. In the field literature, *GoF* functions applied to percentage errors (PE), such as *RMSPE* or *MAPE*, are also referred to as normalized error (NE) measures (i.e. *RMSNE* or *MANE*). In this study we prefer to adopt the “PE” notation, as a normalization can be also applied to a *GoF* function as a whole (see e.g., the *NRMSE*).

It is worth noting that the *NRMSE*($\cdot$) in [Tables 1 and 2](#) has been also referred to as an “absolute” error, as opposed to a “relative” error, i.e., as opposed to *RMSPE*($\cdot$) (see [Kesting and Treiber, 2008](#)). Moreover, *RMSPE*($\cdot$) in [Tables 1 and 2](#) has been also referred to as a “mixed” error measure, since it is a combination of an “absolute” error and a “relative” error.

2.2. Comments about the literature review

Our comprehensive literature review reveals three unsettling points:

- out of 994 articles we have collected and scrutinized from the target journals since 2005, only a small fraction (9%) attempted to calibrate *CF* models. Among these articles:
- an array of *MoPs* was used:
 - 41% calibrated on spacing; 34% on speed, 13% on acceleration, 9% on spacing and speed, and 3% on speed standard deviation.
 - 91% calibrated on a single *MoP*, and 9% on multiple *MoPs*.
- an array of *GoFs* was used: *RMSE* (42%), *RMSPE* (14%), *Theil's U* (20%), *MAE* (9%), *MAPE* (2%), *RMSPE* (7%), and *NRMSE* (6%).

The wide spectrum of calibration settings used in these recent studies, and reported in [Table 1 and 2](#), indicates that the state-of-the-art of *CF* model calibration has been plagued with confusion and contradictions. Even worse, many of these calibration settings are wrong, as they are not structurally able to reproduce the observed car-following dynamics (see [Punzo et al. 2012](#)).

Sometimes a *CF* model is calibrated on a traffic characteristic that is a nonlinear transformation of basic *CF* model outputs, i.e. of spacing, speed and acceleration (e.g., [Tian et al., 2016](#)). Calibrating a model on a nonlinear transformation of an output, e.g. the speed standard deviation, means calibrating a nonlinear transformation of the model itself. As a consequence, when a model is calibrated on such a measure transformation, though it will reproduce the intended characteristic at its best, resulting calibrated *CF* dynamics will be non-optimal.

Remark 4. The only *MoPs* that shall be applied in the *CF* model calibration problem in Eq. (1) are s , v and a , or their linear combinations.

Leaving aside unsuitable or sub-optimal calibration settings, the wide spectrum of settings emerging from the literature is in itself a hindrance to scientific advancements in the field. In fact, it undermines transparency and reproducibility, making it impossible to quantitatively compare results from different studies. In contrast, had all these studies adopted the same calibration setting, it would be possible to compare different models from studies using the same data, or compare performances of a model against different datasets. By doing so, weak or powerful models, and corresponding developments, would be easily identified and distinguished, which would enable researchers to easily reproduce results reported by other researchers, and choose the most promising research path while abandoning the misleading ones. Actually, the benefits of having the same calibration setting appear even before a study is published. At the paper review stage, the same calibration setting can help in achieving a fairer and impartial evaluation of studies, while saving the authors' and reviewers' time.

In the next two sections, therefore, we will present theoretical and experimental bases to draw a sound guideline on what calibration settings shall or shall not be used.

3. The solved case of *MoP*

Looking at [Table 1](#), one may conclude that there is still no consensus on what *MoP* should be used in a *CF* model calibration setting. In truth, this long-standing methodological problem in the car-following theory was solved in a recent paper, [Punzo and Montanino \(2016\)](#).

Depending on the *MoP* used in calibration, the resulting set of optimal parameter values for a *CF* model will be different (again, since we do not have a perfect *CF* model). If we calibrate on speed, we will reach optimality on speed, but resulting spacing will be non-optimal (optimality on spacing is reached when calibrating on spacing, of course). In other words, we have contrasting optimization objectives, which inevitably leads to the fact that the optimality on one *MoP* is detrimental to the optimality on another *MoP*. Thus, if we calibrate on speed, we cannot reduce the error on spacing without at the same time worsening the error on speed, and vice versa. This property is also known as Pareto-efficiency and the two optimal solutions are both Pareto-efficient (Iancu and Trichakis, 2014).

Notwithstanding, the two optimal solutions are not equally desirable from a modelling point of view. In fact, we can quantify the degradation of one measure when the model is calibrated on the other, and vice versa. Between the two, preference will go to the measure producing the least error in the other measure, relative to the minimum obtainable. In fact, the modeler's objective is to obtain a model which is able to consistently reproduce both speed and spacing.

In Punzo and Montanino (2016), a sound physical interpretation of this issue was provided based on mathematical developments, i.e. why a calibration on spacing generally yields a relative error on speed which is smaller than the error on spacing caused by a calibration on speed.

The interpretation is based on Eq. (2) and Eq. (3). In the equations, specifications of the objective functions are given for a model calibration on speed and spacing, respectively (the adopted *GoF* is the Sum of Squared Errors, *SSE*):

$$\min\{SSE^V\} = \min\left\{\sum_{k=1}^N (\epsilon_k^V)^2\right\} \quad (2)$$

$$\min\{SSE^S\} = \min\left\{\sum_{k=1}^N (\epsilon_k^S)^2\right\} = \min\left\{\sum_{k=1}^N (\epsilon_k^V)^2 + (k^*(\epsilon_k^V)^2)[N] + 2\sum_{k=1}^{N-1} \epsilon_k^V \cdot \sum_{i=k+1}^N (N-i+1) \cdot \epsilon_i^V\right\} \quad (3)$$

where ϵ_k^V and ϵ_k^S are, respectively, the speed and spacing error at time k , and are a function of model parameters.

Eq. (3) clarifies that a minimization of *spacing errors* is equivalent to a simultaneous minimization and optimal time allocation of *speed errors*. Here “optimal time allocation” is intended as the allocation of speed errors producing the least error in spacing.

Thus, a calibration on spacing is *optimal in spacing* and *suboptimal in speed*.

On the contrary, a trajectory resulting from a calibration on speed is *optimal in speed* but *indeterminate in spacing* (i.e., we can have the same error in speeds with very different spacing profiles and very different spacing errors); see Eq. (2).

To quantify the loss imposed by one measure on the other, Punzo and Montanino (2016) proposed a relative error measure, as in Punzo and Simonelli (2005). This relative error, RE_i , measures the degradation in a *MoP* “ i ” – relative to its optimum – of a model calibrated on a *MoP* “ j ”. For instance, in case a *CF* model is calibrated on speed, the performance degradation in spacing can be calculated as:

$$RE_s(F(\beta_{RMSE(v)}^*)) = \frac{RMSE_s(F(\beta_{RMSE(v)}^*)) - RMSE_s(F(\beta_{RMSE(s)}^*))}{RMSE_s(F(\beta_{RMSE(s)}^*))} \quad (4)$$

where:

- $\beta_{RMSE(i)}^*$ is the vector of optimal parameter values resulting from a calibration using $RMSE(i)$ as objective function, with $i = s, v$, i.e. spacing or speed;
- $F(\beta_{RMSE(i)}^*)$ is the output of a model simulation with the optimal parameter values $\beta_{RMSE(i)}^*$, see Eq. (1);
- $RMSE_s(F(\beta_{RMSE(i)}^*))$ is the error on spacing given by a model calibrated on “ i ”.

The application of Eq. (4) to thousands of calibrated trajectories allowed a quantification of the relative degradation. The two distributions of RE_s and RE_v confirmed that, in general, a calibration on spacing causes a sensibly smaller degradation in speed than the opposite.

Since this result is due to the integral relationship between speed and spacing, as explained in Punzo and Montanino (2016), we can easily extend the result to acceleration and speed. We can therefore conclude that, when a single *MoP* is adopted in a *CF* calibration objective function (i.e., a single-objective optimisation), spacing is preferable to speed and acceleration.

4. Solving the case of *GoF*

Once the *MoP* case is solved as discussed above, the following questions still need an answer:

1. Given that spacing is preferable as *MoP* in *CF* calibration, what is the best *GoF* function if a single-objective optimization is carried out?
2. Can we improve calibration results through a multi-objective optimization?
3. What is the best *GoF* in a multi-objective optimization setting?

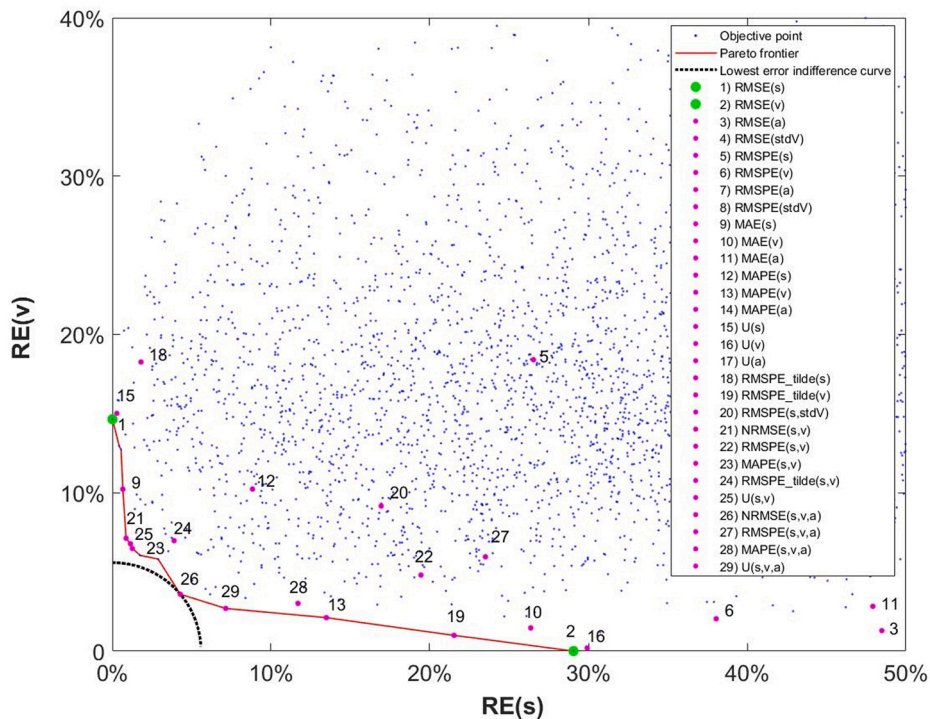


Fig. 1. Results of a calibration experiment in terms of relative errors on spacing and speed. Magenta and green objective points represent the result of calibrations on the listed *GoFs(MoPs)*. Blue points represent model errors corresponding to quasi-randomly generated parameter vectors. The red line is the Pareto frontier. The dashed black line is the indifference curve closest to the origin, i.e., to the utopic point (some *GoF(MoP)* points lie outside the plot limits). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In this section we will address these crucial questions concerning *CF* calibration settings.

A comprehensive study on the topic was [Punzo et al. \(2012\)](#). In that study, the problem of choosing a proper setting was investigated using synthetic trajectories generated from a set of pre-defined parameter values. As mentioned in [Section 2](#), the study objective was to verify what calibration setting was able to rediscover the pre-defined parameter values. On the one hand, this fair and unequivocal testing environment allowed inappropriate settings to be identified, including ineffective algorithms and *GoFs*. On the other hand, since synthetic trajectories were used, it was impossible to provide a clear answer to what *GoF/MoP* combination is preferable. When using synthetic trajectories, a multi-objective problem does not arise. For the values of parameters used to generate the synthetic trajectories a model provides a zero error for all *MoPs*. Thus, there is no trade-off on calibration errors among different *MoPs*.

Therefore, to answer the previous questions, we have calibrated models on real vehicle trajectories. Each combination of *GoF* and *MoPs* identified in the field literature has been tested (see [Table 2](#)),² including two combinations never tested before i.e. *NRMSE(s, v)* and *NRMSE(s, v, a)*. Four sets of trajectory data, which are significantly different from each other in terms of individual vehicle dynamics and empirical/experimental contexts, have been applied. Specifically, three datasets contain human-driven vehicle trajectories, while one contains automated vehicle trajectories. Four car-following models, have been calibrated for each dataset and each *GoF/MoP* combination. These are the IDM ([Treiber et al., 2000](#)), the Gipps' Model ([Gipps, 1981](#)), the FVDM ([Jiang et al., 2001](#)) with either a Constant Time Headway (CTH) or a nonlinear (SIGMOID) spacing policy (the last two are also widely applied as ACC algorithms).

The four models have been selected since they are substantially different from each other in terms of their assumptions and dynamical behaviours (in the goal of ensuring the generality of our results).³ In addition, their properties have been deeply investigated in previous literature (see e.g., [Wilson, 2001](#); [Punzo and Tripodi, 2007](#); [Ward and Wilson, 2011](#); [Treiber and Kesting, 2011](#); [Ciuffo et al., 2012](#); [Punzo et al., 2015](#); [Zhang and Orosz, 2016](#); [Sun et al., 2018, 2020](#); [Montanino et al., 2021](#); [Montanino and Punzo,](#)

² Since the response surfaces produced by *NRMSE(s)*, *NRMSE(v)*, and *NRMSE(a)* functions are linear transformations of the response surfaces produced by the corresponding *RMSE* function, they were not tested in the calibration experiments, i.e. 29 objective functions have been tested in this study.

³ Stochastic *CF* models (i.e. models which have a stochastic term in the formulation, see e.g., [Tian et al. 2021](#)) have not been tested despite their expected higher quality, since their stochastic term might have injected noise into analysis. As mentioned, in the aim of comparing *CF* calibration settings, diversity of models has been considered as the main criterion for model selection. In addition, stochastic models are not best suited for simulating automated vehicles, whose algorithms are expected to be deterministic (see the AstaZero dataset in [Section 4.2](#)).

2021).

4.1. Methodology

As discussed before, *CF* model calibration is a multi-objective optimization problem. Solutions β^* to this problem, also known as Pareto-efficient, yields objective points $z = f(\beta^*)$ located on the so-called Pareto frontier. A solution yielding an objective point on the frontier is also called non-dominated, since it is not possible to find any other solution performing better in all objectives (see Fig. 1).

As mentioned in Section 3, not all Pareto-efficient solutions are equivalent. Usually in micro-economics, “indifference curves” are introduced to compare solutions. In the objectives space $f(\beta) \subseteq \mathbb{R}^n$, each curve is the locus of the objective points $f(\beta)$ that share the same utility for a consumer.

In our case, if we assume that all objectives have the same importance for the analyst (i.e. errors in spacing, speed and acceleration are equally weighed), an indifference curve is given by all objective points having the same distance from the so-called utopic point z^* . This point is the ideal objective point obtained by adopting the minimum achievable error for each objective. In case of two objectives/*MoPs*, e.g., speed and spacing, an indifference curve is any circle in the objectives plane centred in the utopic point; in case of three objectives, i.e., s, v, a , we will have a sphere.

All that said, the most preferable solution among the Pareto-efficient ones is the solution whose objective point lies on the indifference curve closest to the utopic point. That is, the Pareto-optimal solution β^* whose objective point $f(\beta^*)$ has the minimum distance from the utopic point among all other Pareto-optimal solutions. See Fig. 1.

In the figure, the utopic point coincides with the axes origin since each objective is a relative error i.e., normalized over the minimum. In general, in the calibration of a *CF* model, if we assume to measure the error on each objective “ i ” by the *RMSE*, i.e. $f(\beta) = \text{RMSE}(F(\beta))$, the coordinates of the utopic point, z^* , are:

$$z_i^* = \min_{\beta} \{ \text{RMSE}_i(F(\beta)) \}, \text{ with } i = s, v, a \quad (5)$$

where:

- β is a vector of model parameters values;
- $F(\beta)$ is the output of a *CF* model simulation using the parameter vector β , see Eq. (1).

Thus, for every objective point $f(\beta)$ we can calculate the relative error with respect to the utopic point (see points in Fig. 1).⁴ In particular, given the optimal parameter vector $\beta_{GoF(MoP)}^*$ – resulting from a model calibration with an objective function *GoF(MoP)* in Table 2 – the errors of the optimal solution $\beta_{GoF(MoP)}^*$ relative to the utopic point are:

$$RE_i(F(\beta_{GoF(MoP)}^*)) = \frac{\text{RMSE}_i(F(\beta_{GoF(MoP)}^*)) - z_i^*}{z_i^*}, \text{ with } i = s, v, a \quad (6)$$

Eq. (6) is a generalization of Eq. (4) to any objective function *GoF(MoP)*. In Fig. 1, magenta points represent the relative errors on spacing and speed given by model calibrations on the 29 objective functions in Table 2. It is worth noting that only 10 out of the 29 solutions are Pareto-efficient (i.e. lie on the frontier). Among those, the *NRMSE*(s, v, a) is identified as the most preferable since it lies on the indifference curve closest to the utopic point.

Green points on the axes, that are points of coordinates $(0, RE_v(F(\beta_{RMSE(s)}^*)))$ and $(RE_s(F(\beta_{RMSE(v)}^*)), 0)$, represent the relative errors of calibrations using, respectively, *RMSE*(s) and *RMSE*(v) as the objective functions. It is worth recalling here that, in Punzo and Montanino, 2016, a physical explanation, based on mathematical developments, was given about the reason why the relative error on speed of a model calibrated on spacing is almost always lower than the relative error on spacing of a model calibrated on speed. This condition is expressed by the following relationship (see Fig. 1):

$$RE_v(F(\beta_{RMSE(s)}^*)) < RE_s(F(\beta_{RMSE(v)}^*)) \quad (7)$$

In the current framework, Eq. (7) means that the objective point of a calibration on spacing results closer to the utopic point than that of a calibration on speed. In this study, Eq. (7) is verified in 93.5% of calibrations.

Objective points on the circle closest to the origin, are those with the shortest Euclidean distance from the origin. Sometimes, relative errors in an objective (typically acceleration), are one or more order of magnitude higher than those in the other objectives (see e.g., Table 3 in Section 4.4). In this case, the ranking based on the Euclidean distance is mainly driven by the objective with the highest errors (because of the error squaring), which is contrary to the goal of having a “fair” solution with respect to all objectives, i.e., that seeks a minimization of all objectives, consistently. For this reason, as a distance metric from the utopic point, we propose to use the sum of the three relative errors, that is the sum of the coordinates of the objective point. This metric is adopted hereafter in order to rank Pareto-efficient solutions.

⁴ Relative (i.e., normalized) errors allow errors on different objectives (s, v, a) to be compared.

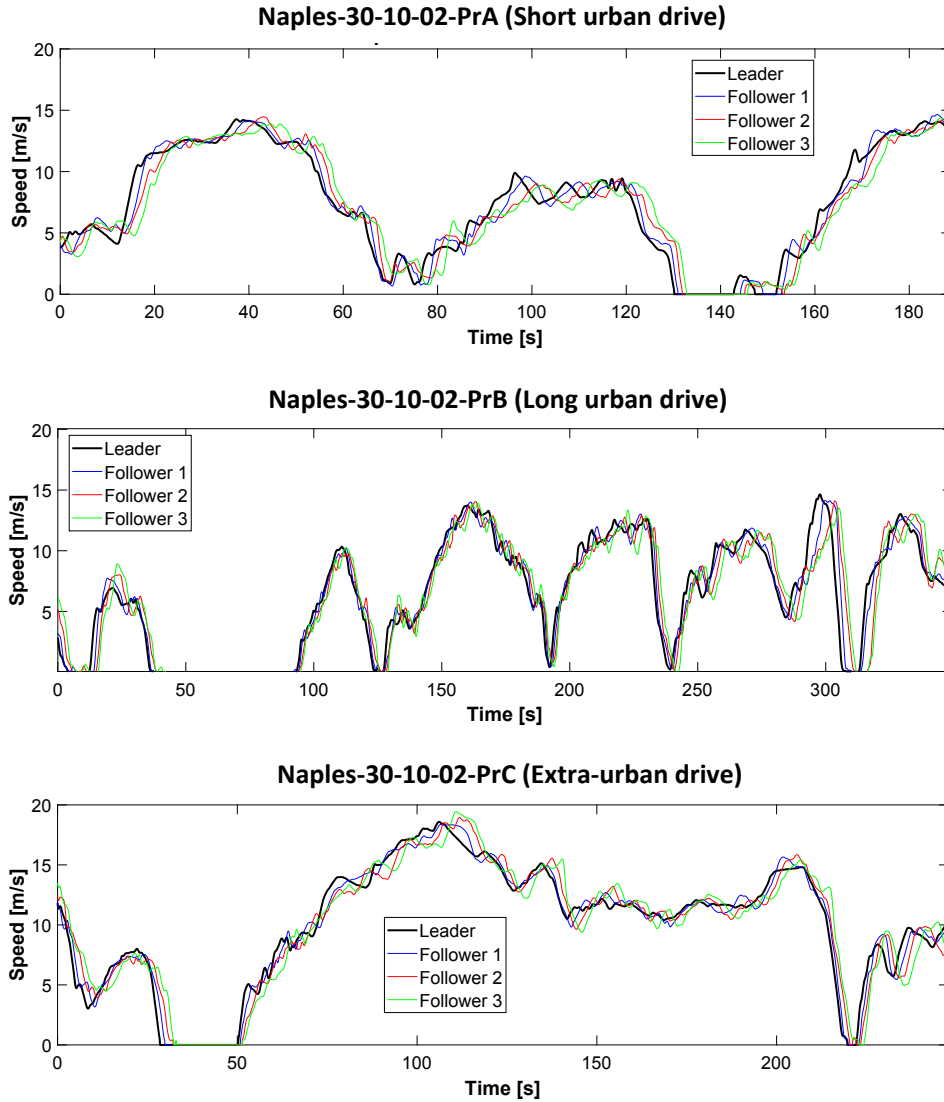


Fig. 2. Human-driven car-following trajectory data from the Naples' field experiment (Punzo and Simonelli, 2005; Punzo et al., 2005).

4.2. Datasets

Four different datasets, including both automated and human-driven vehicles trajectories, have been used in our *CF* calibration experiments, and all trajectories have been pre-processed in order to filter measurement errors. It is worth noting that measurement errors of the same magnitude of those in the adopted datasets, are not expected to impact on *CF* model calibration results (see Montanino and Punzo, 2015, where calibration results on raw and “reconstructed” NGSIM trajectories were compared). Thus, we expect that analysis results are not affected by residual measurement errors.

The first set of trajectories applied here, was collected in three car-following experiments in Naples (Italy) (Punzo and Simonelli, 2005). These experiments were performed with four human-driven vehicles, i.e. a leader and 3 followers, in a platoon along urban and extra-urban roads under different traffic conditions. Each vehicle was equipped with a kinematic differential GPS receiver that recorded the vehicle position at 10 Hz. More details on the dataset and filtering procedure applied to filter raw data, are reported in Punzo and Simonelli (2005) and Punzo et al. (2005). Vehicle speed profiles are depicted in Fig. 2.

The second dataset is a selection of (human-driven) vehicle trajectories from the reconstructed NGSIM I80-1 dataset (Montanino and Punzo, 2015). The most complete trajectories were selected regarding acceleration, cruising, deceleration, following and standstill regimes (Sharma et al., 2019a). The twelve selected trajectories are depicted in Fig. 3.

The third dataset is a selection of human-driven vehicle trajectories from 25-vehicle platooning experiments on a three-lane sub-urban road in Hefei City (Jiang et al., 2015). Each vehicle was equipped with a high-precision differential GPS receiver that recorded the vehicle position at 10 Hz. In each experiment, the leader vehicle was instructed to execute a pre-determined driving cycle. In

Selected NGSIM I80-1 trajectories

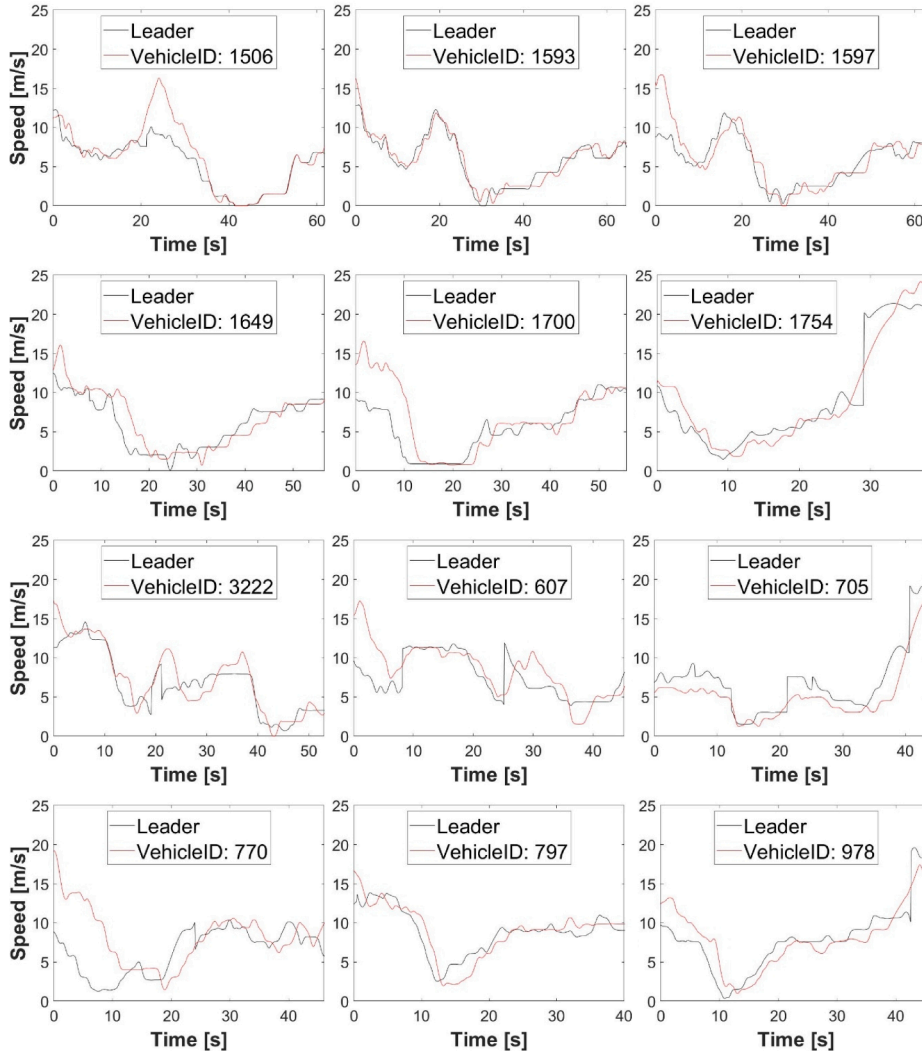


Fig. 3. Selected human-driven car-following trajectory data from the NGSIM I80-1 “reconstructed” dataset (Montanino and Punzo, 2015). Jumps in the leader speed profiles are due to lane-changings (either of the leader or the follower).

addition, at the beginning and at the end of the road, vehicles were forced to make a U-turn. Those trajectory portions were discarded, since their vehicles dynamics were not in car-following. Trajectories were selected according to completeness requirements (Sharma et al., 2019a), as in the second dataset, and filtered through steps 1 and 2 from the “reconstruction” procedure in Montanino and Punzo (2015). Resulting trajectories are shown in Fig. 4.

The fourth dataset contains trajectories made by ACC vehicles collected by the European Commission Joint Research Centre from the field experiments carried out on the rural road of the AstaZero proving ground in Sweden (Makridis et al., 2020). The experiments involved five high-end vehicles (Audi A8, Tesla Model 3, BMW X5, Audi A6 and Mercedes-Benz A 250) driving in a platoon and longitudinally controlled by vehicle ACC systems. Data acquisition was accomplished by the RT-Range S multiple target ADAS measurements solution, which provided vehicle trajectory data at 10 Hz. Two experiments have been selected for this study, whose trajectories are shown in Fig. 5.

4.3. Models

Concerning the models applied, formulations of the IDM, Gipps’ model and FVDM are reported in Eqs. (8), (9) and (10), respectively:

Selected Hefei trajectories

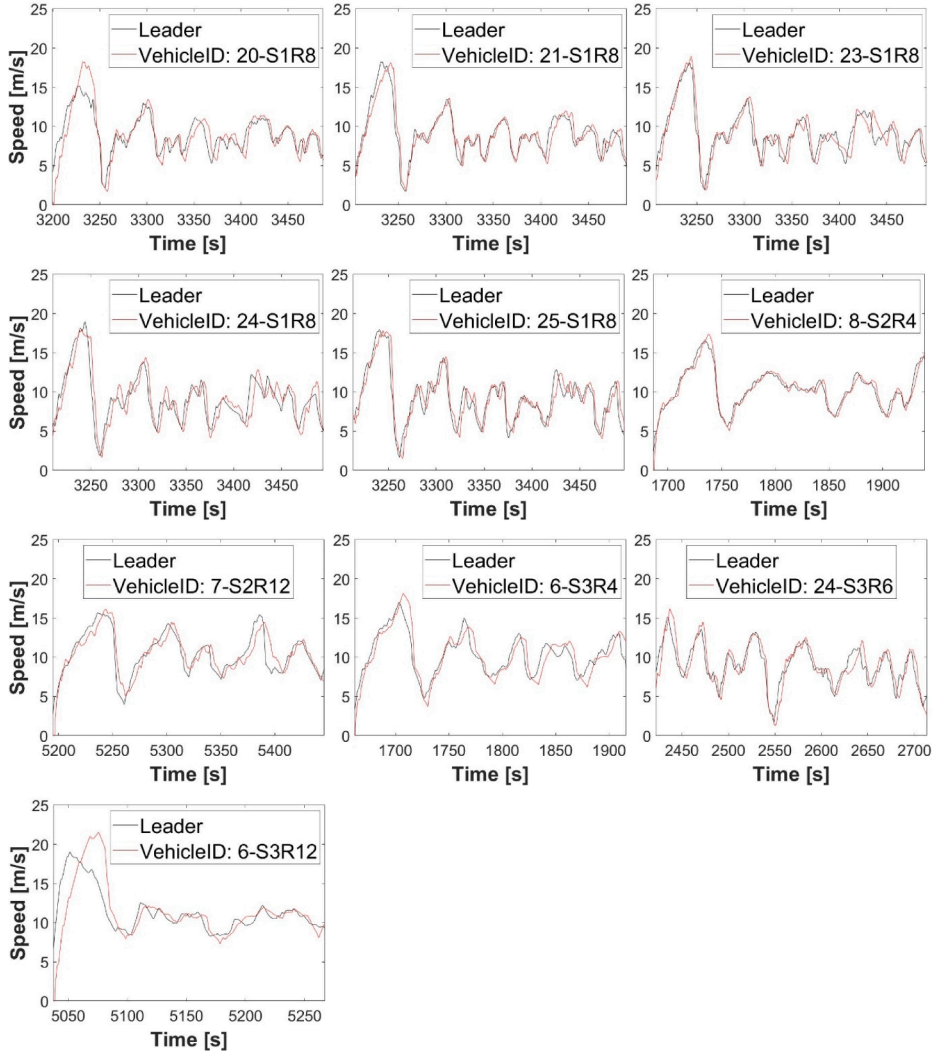


Fig. 4. Selected human-driven car-following trajectory data from the Hefei field experiment (Jiang et al., 2015).

$$a_n(t) = a_{max,n} \cdot \left[1 - \left(\frac{v_n(t)}{V_{max,n}} \right)^{\delta_n} - \left(\frac{s_{0,n} + T_n \cdot v_n(t) - \frac{v_n(t) \cdot \Delta v_n(t)}{2 \cdot \sqrt{a_{max,n} \cdot b_n}}}{s_n(t)} \right)^2 \right] \quad (8)$$

$$a_n(t) = \frac{\min \left\{ \begin{aligned} &v_n(t) + 2.5 \cdot a_{max,n} \cdot \tau_n \cdot \left(1 - \frac{v_n(t)}{V_{max,n}} \right) \cdot \sqrt{0.025 + \frac{v_n(t)}{V_{max,n}}}, \\ &-b_n \cdot \left(\frac{\tau_n}{2} + \theta_n \right) + \sqrt{b_n^2 \cdot \left(\frac{\tau_n}{2} + \theta_n \right)^2 + b_n \cdot \left[2 \cdot (s_n(t) - s_{0,n}) - \tau_n \cdot v_n(t) + \frac{v_{n-1}(t)^2}{\widehat{b_n}} \right]} \end{aligned} \right\}}{\tau_n} - \frac{v_n(t)}{\tau_n} \quad (9)$$

$$a_n(t) = K_{1,n} \cdot [V(s_n(t)) - v_n(t)] + K_{2,n} \cdot \Delta v_n(t) \quad (10)$$

where the function $V(s_n(t))$ for the CTH and SIGMOID spacing policies are defined in Eqs. (11a) and (11b), respectively. Formulations in Eqs. (11a) and (11b) have been widely adopted with Eq. (10) as spacing policies for ACC (e.g. Di Bernardo et al., 2015; Zhang and Orosz, 2016; Zhang et al., 2018; Di Vaio et al., 2019):

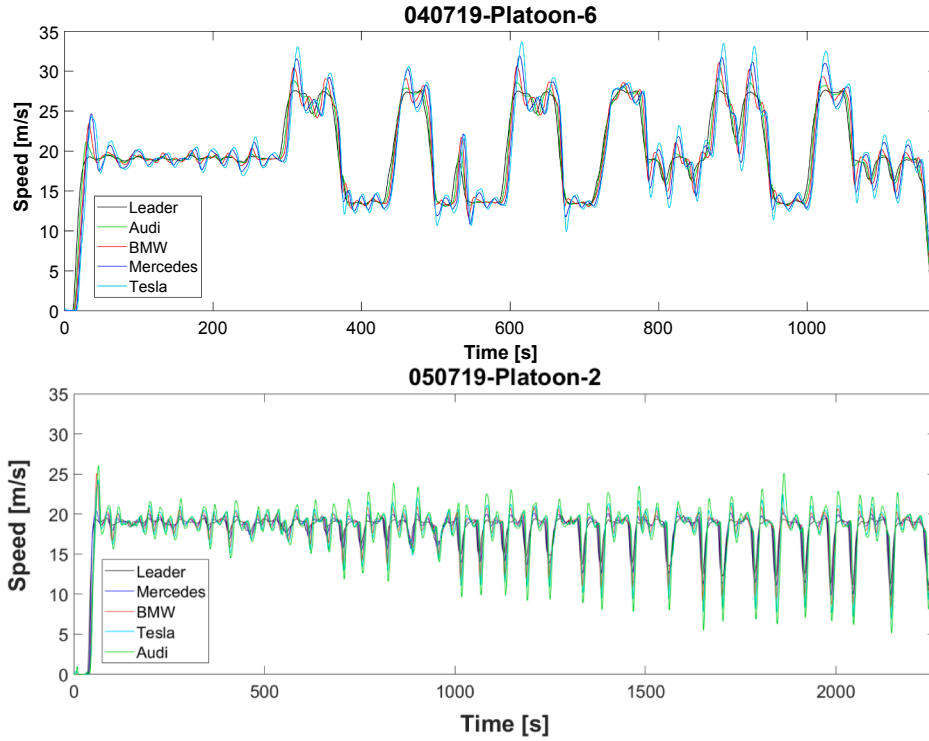


Fig. 5. ACC car-following trajectory data from the AstaZero field experiments (Makridis et al., 2020).

$$V(s_n) = \begin{cases} 0 & \text{if } s_n \leq s_{0,n} \\ \frac{s_n - s_{0,n}}{T_n} & \text{if } s_n \leq s_{0,n} + T_n \cdot V_{\max,n} \\ V_{\max,n} & \text{elsewhere} \end{cases} \quad (11a)$$

$$V(s_n) = \begin{cases} 0 & \text{if } s_n \leq s_{0,n} \\ \frac{V_{\max,n}}{2} \left[1 - \cos\left(\pi \frac{s_n - s_{0,n}}{T_n \cdot V_{\max,n}}\right) \right] & \text{if } s_n \leq s_{0,n} + T_n \cdot V_{\max,n} \\ V_{\max,n} & \text{elsewhere} \end{cases} \quad (11b)$$

In Eqs. (8)–(11), $x_n(t)$, $v_n(t)$ and $a_n(t)$ are a follower vehicle position, speed and acceleration at time t , respectively; Δx_n and $\Delta v_n(t)$ are the inter-vehicle spacing and speed difference from the leader vehicle; l_{n-1} is the leader vehicle length; and $s_n(t) = \Delta x_n - l_{n-1}$ is the net inter-vehicle spacing. Model parameters are in bold.

A ballistic integration scheme, as shown in Eq. (12), with an integration step Δt of 0.1 s (the time-resolution of measured data) has been adopted to update vehicle speed and position (Treiber et al., 2006; Treiber and Kanagaraj, 2015).

$$v_n(t + \Delta t) = v_n(t) + a_n(t) \cdot \Delta t \quad (12a)$$

$$x_n(t + \Delta t) = x_n(t) + \frac{v_n(t + \Delta t) + v_n(t)}{2} \cdot \Delta t \quad (12b)$$

The same integration scheme has been also adopted to obtain positions and accelerations from the measured speed profiles in Figs. 2–5 (see Remark 1).

4.4. Experiments and results

A total of 4524 calibration experiments (4 models $\times$ 39 vehicle trajectories $\times$ 29 objective functions) were run on the high-performance computing grid SCoPE of the University of Naples Federico II. The grid consists of 304 Dell blade M600 computing nodes, each one equipped with 4 dual-cores Xeon processors @ 2.33 GHz. The average computing time per calibration experiment was 6.2 min.

The genetic algorithm coded in Matlab 2019® was used in calibration. A population size equal to 100 allowed the algorithm to likely converge to the global optimum solution (Punzo et al., 2012). In addition, 100 replications of some calibration experiments (10

Table 3

Comparison of Gipps' model calibration errors for the Naples and AstaZero datasets.

	GIPPS (Naples)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	NRMSE(s,v,a)	1.75	0.48	0.52	4.3%	2.9%	1.5%	8.8%
	U(s,v,a)	1.75	0.49	0.52	4.8%	3.6%	1.6%	9.9%
	NRMSE(s,v)	1.68	0.51	0.54	0.1%	8.2%	6.3%	14.6%
	U(s,v)	1.69	0.50	0.54	1.0%	7.5%	6.2%	14.8%
	MAPE(s,v)	1.69	0.51	0.54	1.2%	8.3%	6.0%	15.4%
	RMSE(s)	1.67	0.51	0.55	0.0%	9.1%	7.2%	16.2%
	U(s)	1.68	0.52	0.55	0.5%	9.9%	7.8%	18.3%
	MAPE(s,v,a)	1.87	0.50	0.52	11.7%	6.2%	3.0%	20.9%
	RMSPE_tilde(s,v)	1.74	0.52	0.55	3.9%	10.0%	8.9%	22.8%
	MAE(s)	1.72	0.53	0.57	2.7%	12.7%	11.3%	26.7%
	RMSPE_tilde(s)	1.74	0.53	0.57	4.0%	12.7%	11.2%	28.0%
	RMSPE(s,v,a)	2.02	0.52	0.53	20.7%	10.0%	3.6%	34.4%
	RMSPE(s)	1.95	0.53	0.57	16.4%	13.3%	11.6%	41.3%
	MAPE(v)	2.33	0.48	0.52	39.2%	1.7%	2.3%	43.2%
	RMSPE(s,v)	1.99	0.54	0.57	19.1%	15.8%	11.4%	46.2%
	MAPE(s)	1.95	0.55	0.58	16.6%	16.9%	12.9%	46.5%
	RMSPE(s,stdV)	2.00	0.54	0.57	19.7%	15.8%	12.6%	48.2%
	RMSPE_tilde(v)	2.56	0.48	0.52	52.7%	1.7%	1.8%	56.2%
	RMSPE(v)	2.43	0.50	0.53	45.1%	7.6%	4.2%	56.8%
	MAE(v)	2.70	0.47	0.52	61.5%	0.6%	1.3%	63.4%
	U(v)	2.77	0.47	0.51	65.1%	0.0%	1.0%	66.1%
	RMSE(v)	2.80	0.47	0.51	67.2%	0.0%	1.0%	68.2%
	RMSE(a)	3.17	0.49	0.51	89.3%	3.7%	0.0%	93.0%
	MAE(a)	3.52	0.50	0.51	110.0%	6.6%	0.7%	117.3%
	U(a)	3.41	0.56	0.52	103.6%	18.4%	2.0%	124.0%
	RMSPE(stdV)	11.05	0.82	0.57	560.0%	74.7%	12.3%	647.0%
	RMSE(stdV)	13.26	0.84	0.57	692.1%	78.6%	12.2%	782.9%
	MAPE(a)	28.18	1.22	0.58	1582.7%	159.3%	14.3%	1756.3%
	RMSPE(a)	58.24	2.04	0.66	3378.2%	336.0%	29.1%	3743.3%

	GIPPS (AstaZero)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	RMSE(s)	2.15	0.44	0.42	0.0%	12.7%	326.6%	339.3%
	NRMSE(s,v)	2.19	0.45	0.41	2.0%	14.0%	325.2%	341.1%
	NRMSE(s,v,a)	2.19	0.45	0.41	1.7%	14.3%	325.7%	341.7%
	U(s)	2.20	0.45	0.41	2.2%	14.3%	326.1%	342.5%
	MAE(s)	2.20	0.45	0.41	2.4%	14.3%	326.2%	342.9%
	U(s,v)	2.22	0.45	0.42	3.3%	14.1%	327.5%	344.8%
	RMSPE_tilde(s)	2.27	0.46	0.41	5.4%	17.6%	324.3%	347.3%
	U(s,v,a)	2.37	0.49	0.40	10.2%	24.8%	312.9%	347.9%
	RMSPE(s,v)	2.31	0.45	0.42	7.4%	15.5%	327.3%	350.2%
	RMSPE_tilde(s,v)	2.32	0.46	0.42	7.8%	17.1%	327.4%	352.3%
	RMSPE(s)	2.35	0.46	0.41	9.5%	18.5%	324.3%	352.3%
	MAPE(s,v)	2.33	0.46	0.42	8.3%	16.7%	327.5%	352.4%
	MAPE(s)	2.35	0.46	0.41	9.4%	18.5%	324.5%	352.5%
	RMSPE(s,stdV)	2.46	0.45	0.42	14.4%	15.6%	331.8%	361.7%
	RMSE(v)	5.54	0.39	0.43	157.9%	0.0%	345.6%	503.5%
	RMSPE_tilde(v)	5.53	0.40	0.44	157.2%	1.7%	347.2%	506.1%
	U(v)	5.64	0.39	0.43	162.5%	0.3%	345.6%	508.4%
	MAE(v)	5.72	0.39	0.44	166.0%	0.2%	347.8%	514.0%
	MAPE(v)	5.93	0.40	0.44	175.7%	1.7%	349.4%	526.9%
	RMSPE(v)	7.17	0.44	0.43	233.5%	12.3%	343.5%	589.2%
	RMSE(stdV)	17.37	0.98	0.40	707.8%	150.7%	311.7%	1170.2%
	RMSPE(s,v,a)	17.68	1.39	0.30	722.1%	255.3%	205.4%	1182.8%
	RMSPE(stdV)	19.01	0.89	0.41	784.0%	127.5%	319.2%	1230.7%
	MAPE(s,v,a)	40.29	2.09	0.31	1773.6%	433.6%	217.2%	2424.4%
	U(a)	344.64	3.25	0.10	15928.0%	729.5%	5.4%	16662.9%
	MAE(a)	366.18	3.45	0.10	16929.8%	780.7%	0.6%	17711.0%
	RMSE(a)	375.74	3.51	0.10	17374.6%	795.7%	0.0%	18170.2%
	RMSPE(a)	402.76	3.80	0.10	18630.9%	871.0%	4.6%	19506.5%
	MAPE(a)	404.02	3.80	0.10	18689.7%	871.9%	4.4%	19565.9%

Table 4

Comparison of FDVM-CTH model calibration errors for the Naples and AstaZero datasets.

CTH		CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
(Naples)		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	1.99	0.51	0.56	4.2%	5.0%	39.1%	48.2%
	NRMSE(s,v,a)	1.93	0.52	0.57	1.1%	6.3%	42.3%	49.7%
	U(s,v)	1.97	0.52	0.59	3.2%	5.8%	47.2%	56.2%
	NRMSE(s,v)	1.93	0.52	0.60	1.0%	7.1%	48.6%	56.8%
	RMSE(s)	1.91	0.54	0.59	0.0%	10.0%	48.2%	58.1%
	U(s)	1.95	0.53	0.60	1.7%	8.4%	48.9%	58.9%
	RMSPE_tilde(s,v)	2.01	0.54	0.59	5.1%	11.1%	47.2%	63.4%
	MAE(s)	2.01	0.55	0.60	4.9%	12.9%	49.9%	67.7%
	RMSPE_tilde(s)	2.02	0.55	0.60	5.6%	13.1%	50.7%	69.4%
	RMSPE(s,v)	2.20	0.56	0.60	15.2%	14.3%	50.1%	79.5%
	MAPE(s,v)	2.15	0.56	0.62	12.2%	13.9%	53.7%	79.8%
	MAPE(v)	2.63	0.50	0.57	37.6%	2.9%	42.2%	82.7%
	RMSPE(v)	2.60	0.51	0.57	36.1%	5.0%	42.6%	83.7%
	RMSPE_tilde(v)	2.70	0.50	0.57	41.1%	1.7%	41.1%	83.9%
	MAPE(s)	2.19	0.57	0.62	14.7%	16.9%	55.0%	86.6%
	RMSPE(s)	2.30	0.57	0.62	20.4%	17.5%	54.2%	92.0%
	RMSPE(s,stdV)	2.32	0.58	0.62	21.5%	18.8%	53.8%	94.1%
	MAE(v)	2.98	0.49	0.57	55.8%	0.9%	42.8%	99.5%
	U(v)	3.12	0.49	0.57	63.1%	0.4%	41.5%	105.1%
	RMSE(v)	3.20	0.49	0.57	67.5%	0.0%	41.5%	108.9%
	MAPE(s,v,a)	3.08	0.60	0.55	60.8%	23.2%	38.2%	122.2%
	RMSPE(s,v,a)	4.43	0.60	0.54	131.5%	23.6%	34.2%	189.3%
	RMSE(stdV)	3.65	0.61	0.73	90.9%	25.8%	81.9%	198.6%
	RMSPE(stdV)	3.65	0.61	0.73	90.9%	25.8%	81.9%	198.6%
	U(a)	20.61	0.99	0.40	977.6%	102.8%	0.7%	1081.1%
	RMSE(a)	21.20	1.04	0.40	1008.5%	114.0%	0.0%	1122.5%
	RMSPE(a)	22.33	1.06	0.47	1067.7%	116.6%	16.2%	1200.6%
	MAPE(a)	22.44	1.14	0.48	1073.3%	133.2%	20.8%	1227.2%
	MAE(a)	25.68	1.18	0.43	1243.0%	141.4%	8.4%	1392.7%

CTH		CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
(AstaZero)		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	NRMSE(s,v,a)	2.85	0.51	0.42	1.8%	5.4%	41.1%	48.3%
	NRMSE(s,v)	2.83	0.51	0.43	1.2%	5.8%	44.4%	51.4%
	RMSE(s)	2.80	0.51	0.43	0.0%	5.9%	46.5%	52.3%
	RMSPE_tilde(s,v)	2.86	0.51	0.43	2.1%	6.2%	44.6%	52.9%
	RMSPE_tilde(s)	2.85	0.52	0.43	1.7%	6.5%	44.6%	52.9%
	MAE(s)	2.85	0.51	0.43	1.7%	6.2%	46.1%	54.0%
	U(s)	2.85	0.52	0.43	1.6%	7.3%	46.0%	54.9%
	U(s,v)	2.92	0.52	0.42	4.4%	8.0%	42.6%	55.0%
	MAPE(s)	2.91	0.52	0.43	3.8%	7.4%	44.5%	55.6%
	MAPE(s,v)	2.91	0.52	0.43	3.9%	7.3%	44.5%	55.7%
	RMSPE(s,v)	2.96	0.54	0.43	5.8%	11.0%	44.8%	61.7%
	U(s,v,a)	3.20	0.59	0.38	14.3%	21.8%	29.5%	65.6%
	RMSPE(s)	3.00	0.55	0.43	7.2%	14.2%	45.8%	67.3%
	RMSPE(s,stdV)	2.96	0.57	0.45	5.8%	17.6%	53.1%	76.5%
	RMSPE(v)	5.02	0.50	0.42	79.3%	3.6%	43.3%	126.2%
	RMSPE_tilde(v)	5.33	0.49	0.42	90.4%	1.0%	43.7%	135.1%
	RMSE(v)	5.44	0.48	0.42	94.2%	0.0%	42.3%	136.5%
	U(v)	5.48	0.48	0.42	95.7%	0.1%	43.0%	138.9%
	MAPE(v)	5.75	0.49	0.42	105.2%	0.8%	43.1%	149.2%
	MAE(v)	6.27	0.49	0.42	123.7%	0.9%	42.7%	167.4%
	MAPE(s,v,a)	5.07	0.82	0.35	81.0%	68.6%	19.8%	169.4%
	RMSPE(stdV)	5.60	0.57	0.45	99.8%	16.9%	53.7%	170.5%
	RMSPE(s,v,a)	5.92	0.80	0.35	111.3%	64.7%	17.9%	193.9%
	RMSE(stdV)	6.84	0.59	0.45	144.3%	22.6%	52.6%	219.6%
	MAPE(a)	35.82	1.02	0.30	1179.1%	111.7%	1.1%	1291.9%
	RMSPE(a)	35.84	1.02	0.30	1179.5%	111.9%	0.8%	1292.2%
	U(a)	38.18	1.02	0.29	1263.3%	110.5%	0.1%	1373.9%
	RMSE(a)	38.17	1.02	0.29	1262.8%	111.3%	0.0%	1374.1%
	MAE(a)	38.18	1.02	0.30	1263.0%	110.8%	0.4%	1374.3%

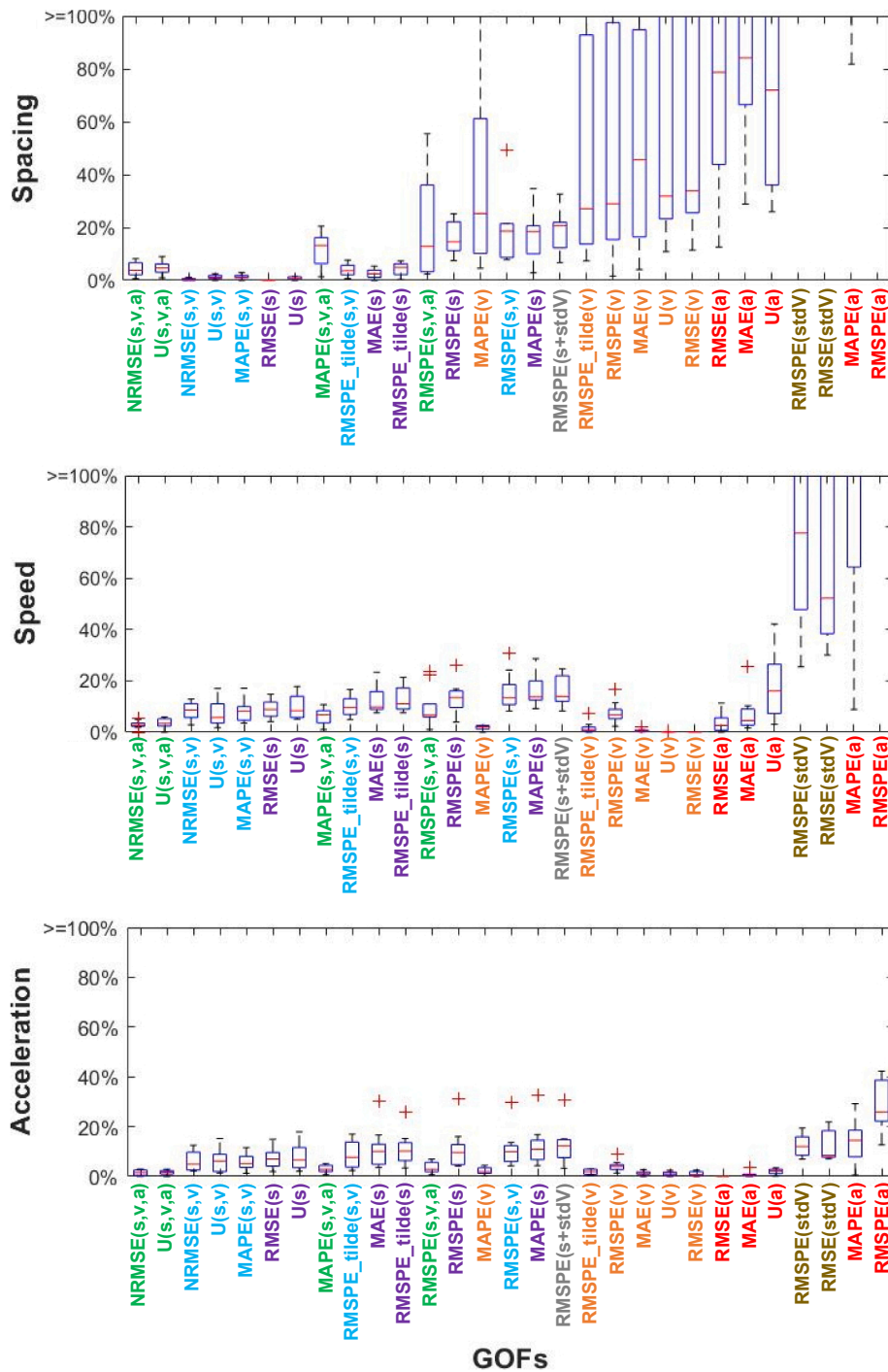


Fig. 6. Relative calibration error box plots of Gipps' model for Naples trajectories. The *GoF(MoP)* functions are sorted according to Table 3. Colours identify *GoFs* with the same *MoP* or combination of *MoPs*. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

randomly chosen experiments out of 4524) were performed to verify that the algorithm converged to the same solution at each calibration replication, both in terms of parameter values and objective function value (the recorded standard deviation among solutions at convergence is equal to $1e-10$). For more details on the configuration of the experiments, please refer to the open-source RStudio codes provided with the paper.

Results about Gipps' model and the FVDM-CTH are presented, respectively, in Tables 3 and 4 for Naples' and AstaZero datasets.

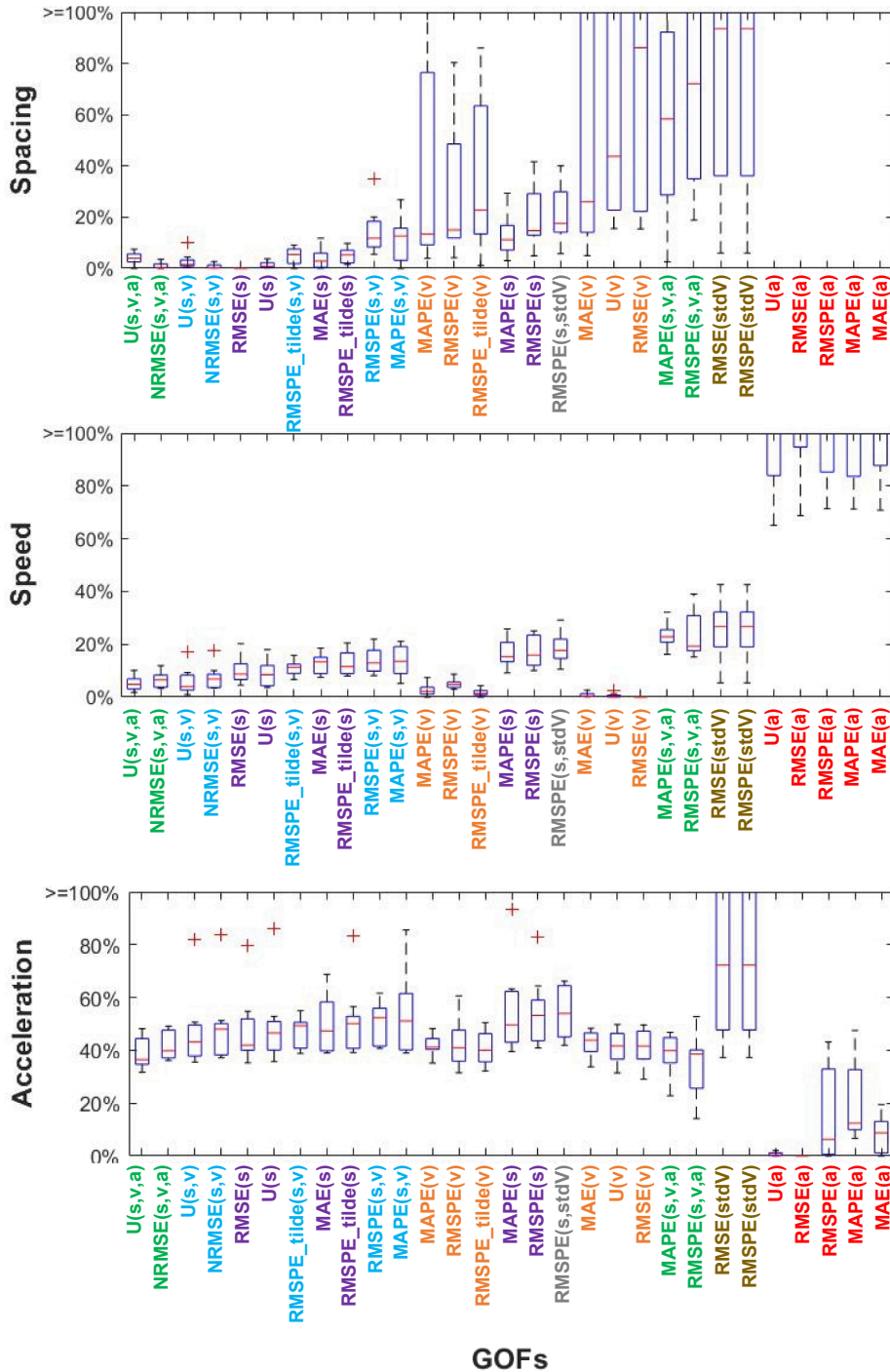


Fig. 7. Relative calibration error box plots of FVDM with CTH spacing policy for Naples trajectories. The $GoF(MoP)$ functions are sorted according to Table 4. Colours identify $GoFs$ with the same MoP or combination of $MoPs$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

These results are deemed to be the most representative as to the variability of results themselves. In fact, Gipps's model and FVDM-CTH are, respectively, the best and the worst performing models over all datasets. Naples dataset is the one showing the least errors for all models among the human-driven vehicles datasets. AstaZero is the only automated vehicles dataset. Results for all the other models and datasets are reported in Appendix C.

In the left half of the tables, calibration errors on spacing, speed and acceleration are shown in terms of $RMSE(s)$, $RMSE(v)$, and

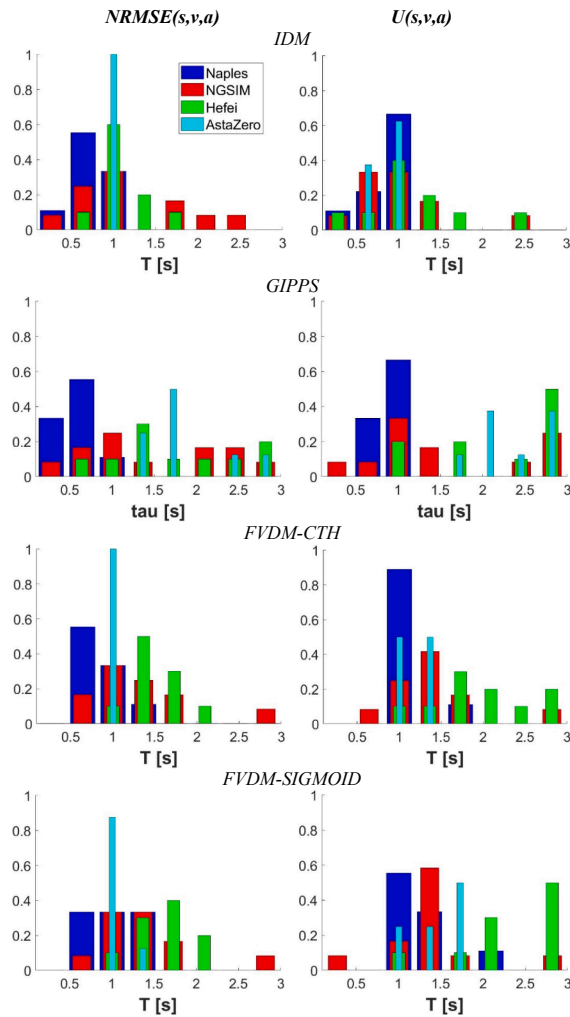


Fig. 8. Histograms of calibrated values of the time headway parameter, by means of $NRMSE(s, v, a)$ and $Theil's U(s, v, a)$, for the four models and the four trajectory datasets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$RMSE(a)$, for the 29 tested $GoF(MoP)$ functions. The reported value is the average over N calibration experiments, where N is the number of trajectories in the dataset.

In the right half of the tables, relative errors as defined in Eq. (6), averaged over the N experiments, are reported for the same objective functions:

$$RE_i\left(F\left(\beta_{GoF(MoP)}^*\right)\right) = \frac{\sum_{k=1}^N RE_i\left(F\left(\beta_{GoF(MoP),k}^*\right)\right)}{N}, \text{ with } i = s, v, a \quad (13)$$

In the rightmost column, the sum of the three relative errors in Eq. (13) is reported:

$$RE\left(F\left(\beta_{GoF(MoP)}^*\right)\right) = \sum_{i=\{s,v,a\}} RE_i\left(F\left(\beta_{GoF(MoP)}^*\right)\right) \quad (14)$$

The 29 tested $GoF(MoP)$ are ranked according to the distance metric in Eq. (14), as anticipated in Section 4.1. Each table cell is coloured according to the percentile of the distribution of each column metric (where the green and red colours correspond to the 0th and 100th percentiles, respectively).

Result analysis shows that these trajectory datasets are very different from each other, since errors in the three objectives sensibly vary across the datasets and it is not possible to identify a pattern of variation.

Concerning the comparison of objective functions, several observations from the results in the tables are worth noting:

- When a single MoP is applied:
 - Regardless of the GoF , the model and the dataset, spacing is preferable as MoP . In fact, inequality in Eq. (7), on average (i.e., see Eq. (13)), and the equivalent one between speed and acceleration hold for every GoF , model and dataset. This is a further experimental confirmation of the results in Punzo and Montanino (2016).
 - Focusing on spacing, regardless of the model and dataset, $GoFs$ which are not based on percentage errors i.e., $RMSE(s)$, $Theil's U(s)$ and $MAE(s)$ are always more preferable than percentage-based $GoFs$ i.e., $RMSPE$ and $MAPE$ (with the only exception of the IDM/Hefei scenario, shown in Appendix C).
- When multiple $MoPs$ are applied:
 - $NRMSE(s, v, a)$ and $Theil's U(s, v, a)$ are the two most preferable Pareto-efficient objective functions in the vast majority of the calibration scenarios. $NRMSE$ is the most robust between the two, since in the ACC dataset for the $FVDM-CTH$, $U(s, v, a)$ is one of the worst performing objective functions among those containing s as MoP .
 - Percentage-based objective functions, i.e., $RMSPE$ and $MAPE$ are less preferable than non-percentage-based ones, i.e., $NRMSE$ and $Theil's U$, either with two $MoPs$, i.e., s, v (with the only exception of the IDM/Hefei scenario), or three $MoPs$, i.e., s, v, a . In addition, in many calibration experiments, percentage-based objective functions did not return Pareto-efficient solutions (in general, percentage-based objective functions are not recommended, since they require thresholds on speed and acceleration, and overweigh lower value measurements in calibration).

It is also worth noting that $NRMSE(s, v, a)$ and $Theil's U(s, v, a)$ have never been used in the literature, to the best of our knowledge. $NRMSE$ was used with a single MoP , i.e. spacing or speed (see e.g., Kesting and Treiber, 2008; Papathanasopoulou and Antoniou, 2015) but, in such a case, it is easy to show that $NRMSE$ is equivalent to $RMSE$ or SSE .

Results in Tables 3 and 4 (plus those in Appendix C) are also useful to compare the model performances, across these different trajectory datasets.

Gipps' model and the IDM turned out to be the best-performing models on human-driven vehicle datasets. In particular, Gipps' model has the lowest error on spacing across all datasets (speed and acceleration errors are comparable to the minimum calibration errors, which are achieved by the IDM). Focusing on the automated vehicles dataset, Gipps' model turned out to have the lowest calibration error on spacing and speed, among the four tested models (the acceleration error is very close to the minimum value, which is achieved by the IDM). It is somewhat surprising that Gipps' model, which is based on an idealized safe human driving behaviour, has been the best model to approximate automated vehicle control logics, outperforming controllers which are customarily applied to simulate ACC vehicles (i.e., $FVDM$ with CTH and $SIGMOID$ spacing policies).

Variances of relative calibration errors across the population of vehicles in Naples' dataset, are visualised using box plots in Figs. 6 and 7, as an example. As shown in these box plots, the variances of the relative calibration errors are significantly different across different objective functions. In general, the objective functions that are based either on speed only, or on acceleration only, display very large variances in their relative errors on spacing. Conversely, the objective functions based on spacing are much more robust, showing small variances of relative errors on speed as well as on acceleration. $NRMSE(s,v,a)$ and $U(s,v,a)$ consistently show small relative calibration errors on all three measures.

Eventually, a comparison of calibrated parameter distributions among the models and among the datasets is presented for the two most preferable Pareto-efficient $GoFs$ (see Fig. 8). The presented distributions should be viewed as another tool to assist us in inspecting the objective functions and models behaviour, and should not be assumed as reference distributions, e.g., for traffic simulation⁵ (concerning the link between microscopic parameters distributions and traffic simulation outputs, the reader can refer to Punzo and

⁵ Reported parameters distributions are representative of each specific dataset and their transferability to other contexts is questionable.

Montanino, 2020).

With the aim of comparing values across all models, the “time headway” parameter has been chosen, since it is the most influential parameter in calibration among those present in all models – the others being the distance at stop s_0 , and the desired speed V_{max} (see Punzo and Ciuffo, 2011; Punzo et al., 2015 as to the IDM and Gipps’ model).

As shown in Fig. 8, for human-driven vehicles, time headway histograms clearly vary among models.⁶ Of course, they vary also across the datasets. A lower average value and a smaller variance is observed for the Naples’ dataset, which was collected from an urban context in tight car-following conditions. Higher average values and larger variances are observed for the NGSIM and the Hefei datasets, which were collected from a highway or highway-like (i.e., no traffic lights) context.

As to the differences between the two objective functions, *Theil’s* $U(s, v, a)$ yields a high number of parameter values falling into a range close to the upper bound as well as more discontinuous histograms. On the contrary, $NRMSE(s, v, a)$ returns more regular histograms.

For automated vehicles, results significantly differ between the two objective functions. As to the $NRMSE(s, v, a)$ two models return time headway values in one class only (i.e. between 0.75 and 1.125 s), with the only Gipps’ model returning values in more than two classes. On the contrary, for *Theil’s* $U(s, v, a)$, histograms clearly vary across the models.

5. A guideline on calibration settings

In light of previous results and in the spirit of the present study, we provide two qualitatively different recommendations.

The first one concerns those calibration settings which are clearly inadequate, and as such should be avoided. This recommendation is deemed to be definitive according to the epistemological assumptions of this study (see the Introduction).

The second recommendation concerns the calibration settings which have shown to be the most preferable among the Pareto-efficient ones. Despite the fact that the study spans over four different models and four different datasets, this recommendation cannot be claimed to be final, since there might be other calibration settings that are not considered here but would perform better than those recommended (in these scenarios and in others).

1) Unsuitable settings in CF model calibration:

- If one single MoP is adopted in the objective function, *speed* and *acceleration* should be avoided. *Spacing* should be used instead, as it is more preferable (see Eq. (7)).
- If two or more MoPs are adopted in the objective function, percentage-based *GoFs* (i.e., $RMSPE$ and $MAPE$) should be avoided, since either they are not Pareto-efficient or they are less preferable than non-percentage-based objective functions like $NRMSE$ and *Theil’s* U .

2) Pareto-efficient settings for CF model calibration:

- $NRMSE(s, v, a)$ and *Theil’s* $U(s, v, a)$ are the most preferable objective functions among those being Pareto-efficient.
- Since both $NRMSE(\cdot)$ and *Theil’s* $U(\cdot)$ have similar performances whether two or three MoPs are considered (i.e. s, v or s, v, a), if acceleration data are noisy (which often is the case in practice), $NRMSE(s, v)$ or *Theil’s* $U(s, v)$ is preferred.

Based on results about parameter distributions (see Fig. 8 concerning the time headway), we hold a certain preference for $NRMSE(\cdot)$ over *Theil’s* $U(\cdot)$.

Note that a proper acquisition and treatment of data are preliminary to the identification of unsuitable and suitable calibration settings. The following three prerequisites need to be guaranteed (see Section 2 and Remarks 1 and 2):

- when pre-processing trajectory data, care must be taken to avoid error amplification in the differentiation process (noise), and error accumulation in the integration process (bias);
- if several measures are independently gathered through separate devices, these measurements need to be (made) “internally consistent”;
- the integration/differentiation scheme of data needs to be the same as the one adopted in the model simulation.

6. Implications on research transparency and reproducibility

For traffic flow research in general, and CF modelling in particular, a widely agreed CF model calibration setting is a prerequisite for transparency and reproducibility of research. Without adopting the same setting in different studies, it would be impossible to quantitatively and objectively compare different CF models, eventually calibrated on different datasets. By adopting the same setting researchers would be helped in choosing the most promising research paths while abandoning the misleading ones, which could also save reviewers’ time.

As the recommendation on the calibration setting provided herein cannot be final, we are happy to share our methodological

⁶ This might sound weird or counterintuitive, as time headway has a clear physical interpretation. Thus, one might expect calibrated distributions to be similar among models and to the empirical distribution of the quantity, i.e., as directly measured from data. Since calibrated parameters account for the whole epistemic uncertainty (for instance, they also compensate for model inaccuracies or misspecifications), estimated distributions result is inevitably different from the measured one (see discussions in Punzo and Montanino, 2020, and Punzo et al., 2015).

framework and corresponding computer codes with the scientific community. In this way, any researcher can compare any new calibration setting, or model or dataset in a fair and objective manner against several models, settings and datasets, thus substantially speeding up the scientific knowledge advancement.

In fact, transparency and reproducibility are generally regarded to as the gold standard of validating a scientific study. Simply speaking, reproducible research is an extra (but very important) effort from the authors of a publication to share their data, codes, and instructions on how to piece them together for the purpose of enabling a third party to easily and conveniently obtain the identical results reported in the paper (Mesirov 2010; Peng, 2011). Catalyzed by powerful, user-friendly and freely-available tools, reproducible research has been gaining lots of momentum and is at the verge of fundamentally changing how scientific work related to computational analysis should be conducted, documented, disseminated, and maintained (Donoho et al. 2008; Donoho, 2010; Gandrud, 2016; Xie et al. 2018). However, to date the Transportation Research community has not paid much attention to this reproducible research movement (Zheng, 2020).

For the reasons above we hope that reproducibility of research will become a standard also in the Transportation Research field. Meanwhile, this paper is totally reproducible. The source codes for generating the entire paper in RStudio can be downloaded at <http://www.multitude-project.eu/> or <http://www.connectedandautonomoustransport.com/reproducible-research.html>. Introduction to and tools for reproducible research can be found in Zheng (2020).

7. Concluding remarks

We are at a crossroads where it has to be decided how academic work should be conducted, documented and disseminated. The belief that reproducible research is an unstoppable and inevitable future has been gaining a massive following in many disciplines (e.g., biostatistics, signal processing, statistical analysis, etc.). Still, it is not widespread in the Transportation Research community.

By providing clear guidelines for calibrating car-following models, and making this paper totally reproducible, it is the authors' hope that it could not only end the state of confusion and contradictions that has plagued the *CF* modelling literature for so long, but also inspire the start of a wide practice of transparent and reproducible research in *CF* dynamics modelling. A spark that could hopefully spread into other branches of Transportation Research.

In order to provide a guideline on calibration setting we have made an extensive calibration exercise by comparing all the objective functions adopted in the field literature so far, and some new ones. The analysis has been based on four models of car-following dynamics – widely adopted in the fields of both traffic flow theory and control theory – and on four different datasets, including trajectories of human-driven and automated vehicles.

A methodology to compare and rank objective functions in terms of Pareto-efficiency has been developed. The criterion proposed, in particular, gives preference to the objective function providing the least distance from the utopic point. In order to compare results from objective functions with different units of measurement, a relative error with respect to the minimum achievable error over each objective has been proposed.

Based on the analysis of the extensive calibration experiments carried out, two qualitatively different recommendations have been provided.

The first one discourages the use of the objective functions which have consistently shown to be either not Pareto-efficient, or not preferable according to the proposed criterion across all the models and datasets. This recommendation is deemed to be final based on the epistemological assumptions of this study. In fact, these objective functions have no general validity. If one still intends to use them in a specific case, their scenario-specific superiority must be demonstrated against all other objective functions, before use.

The second recommendation is about calibration settings which have shown to be the most preferable according to the proposed criterion, among the Pareto-efficient ones. Despite the fact that the study spans over four different models and four different and diverse datasets, this recommendation cannot be claimed to be final, since there might be other calibration settings, which are not considered here, that could perform better than those recommended. However, aside from new objective functions, an upheaval of the provided ranking is not expected, given the robustness of presented results across the quite wide range of models and datasets used in this study.

Any different model, calibration setting, or trajectory dataset could be plugged-in in the computer codes that we share with the community. Thus, it could be tested in a fair and impartial manner against models, settings and trajectories applied in this study or added by other researchers.

CRedit authorship contribution statement

Vincenzo Punzo: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Supervision, Validation, Writing - original draft, Writing - review & editing. **Zuduo Zheng:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Supervision, Validation, Writing - original draft, Writing - review & editing. **Marcello Montanino:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing.

Table C1

Comparison of calibration errors for the IDM and for the FVDm with SIGMOID spacing policy, calibrated against Naples trajectories.

	IDM (Naples)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\Sigma_i \overline{RE}(i)$
CALIBRATION GOFs	NRMSE(s,v,a)	1.76	0.47	0.50	5.8%	3.9%	2.1%	11.7%
	U(s,v,a)	1.76	0.47	0.50	5.9%	4.7%	2.3%	13.0%
	U(s,v)	1.68	0.49	0.52	1.4%	8.1%	6.3%	15.8%
	NRMSE(s,v)	1.67	0.49	0.52	0.8%	8.7%	6.5%	16.0%
	MAPE(s,v)	1.69	0.49	0.52	2.1%	7.9%	6.0%	16.0%
	MAPE(s,v,a)	1.84	0.47	0.50	10.9%	4.3%	2.4%	17.6%
	RMSPE_tilde(s,v)	1.73	0.49	0.53	4.2%	9.3%	7.6%	21.2%
	RMSPE(s,v,a)	1.91	0.47	0.50	14.9%	4.2%	2.3%	21.4%
	RMSE(s)	1.66	0.51	0.54	0.0%	12.6%	10.1%	22.8%
	U(s)	1.70	0.51	0.54	2.6%	12.8%	9.6%	25.1%
	MAE(s)	1.73	0.50	0.54	4.5%	11.9%	9.1%	25.5%
	RMSPE_tilde(s)	1.75	0.51	0.54	5.3%	13.5%	10.5%	29.3%
	MAPE(v)	2.12	0.46	0.50	27.6%	1.6%	2.5%	31.6%
	RMSPE(s,v)	1.91	0.50	0.54	15.2%	11.0%	9.1%	35.3%
	MAPE(s)	1.91	0.52	0.55	14.8%	15.2%	11.1%	41.1%
	RMSPE(v)	2.22	0.47	0.51	33.8%	3.5%	4.5%	41.8%
	RMSPE(s)	1.94	0.51	0.55	17.1%	14.1%	11.1%	42.3%
	RMSPE(s,stdV)	1.97	0.51	0.55	18.5%	14.1%	11.1%	43.7%
	RMSPE_tilde(v)	2.46	0.45	0.50	48.3%	0.6%	2.0%	50.9%
	MAE(v)	2.54	0.45	0.50	52.7%	0.9%	1.5%	55.1%
	RMSE(v)	2.63	0.45	0.50	58.7%	0.0%	1.4%	60.1%
	U(v)	2.66	0.45	0.50	60.0%	0.0%	1.2%	60.8%
	RMSE(a)	2.77	0.46	0.49	67.1%	2.8%	0.0%	69.9%
	U(a)	2.69	0.48	0.50	62.3%	6.3%	1.4%	70.0%
	MAE(a)	2.85	0.46	0.49	71.7%	1.6%	0.5%	73.8%
	RMSE(stdV)	3.58	0.58	0.58	115.9%	28.9%	18.7%	163.5%
	RMSPE(stdV)	4.62	0.62	0.60	178.2%	38.4%	21.4%	238.0%
	MAPE(a)	5.41	0.54	0.51	225.8%	20.7%	3.9%	250.4%
	RMSPE(a)	244.07	4.18	0.75	14603.2%	827.8%	53.0%	15484.1%

	SIGMOID (Naples)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\Sigma_i \overline{RE}(i)$
CALIBRATION GOFs	NRMSE(s,v,a)	1.92	0.52	0.57	0.2%	5.7%	41.5%	47.4%
	U(s,v,a)	2.01	0.52	0.56	4.9%	5.0%	39.1%	49.0%
	NRMSE(s,v)	1.97	0.53	0.59	2.7%	7.5%	46.4%	56.6%
	RMSPE_tilde(s,v)	2.01	0.54	0.58	4.8%	9.4%	44.9%	59.1%
	U(s,v)	2.00	0.53	0.60	4.1%	7.0%	48.6%	59.7%
	RMSE(s)	1.92	0.54	0.61	0.0%	9.1%	51.0%	60.1%
	RMSPE_tilde(s)	2.03	0.55	0.59	5.8%	11.0%	47.6%	64.4%
	MAE(s)	2.05	0.55	0.61	7.1%	11.0%	51.4%	69.4%
	U(s)	2.00	0.55	0.66	4.6%	11.6%	63.3%	79.4%
	MAPE(s,v)	2.21	0.57	0.64	15.2%	15.3%	58.3%	88.8%
	RMSPE(s,v)	2.44	0.58	0.61	27.3%	17.4%	52.2%	96.8%
	MAE(v)	3.18	0.50	0.56	66.1%	0.7%	38.8%	105.7%
	RMSPE(v)	3.06	0.52	0.58	59.8%	5.0%	43.0%	107.8%
	U(v)	3.31	0.49	0.56	72.6%	0.0%	38.1%	110.7%
	MAPE(s)	2.25	0.59	0.70	17.5%	20.4%	73.4%	111.4%
	RMSPE(s)	2.51	0.60	0.65	30.7%	21.1%	60.4%	112.2%
	RMSE(v)	3.38	0.49	0.56	76.5%	0.0%	38.0%	114.5%
	RMSPE_tilde(v)	3.41	0.50	0.56	77.8%	1.4%	38.7%	117.9%
	RMSPE(s,stdV)	2.67	0.60	0.65	39.2%	22.3%	61.5%	123.0%
	MAPE(v)	3.45	0.51	0.57	80.2%	3.4%	40.8%	124.4%
	MAPE(s,v,a)	3.19	0.61	0.58	66.5%	24.7%	44.0%	135.2%
	RMSPE(s,v,a)	3.82	0.62	0.62	99.5%	25.0%	53.8%	178.3%
	RMSE(stdV)	4.98	0.61	0.64	160.0%	22.7%	58.9%	241.6%
	RMSPE(stdV)	4.98	0.61	0.64	160.0%	22.7%	58.9%	241.6%
	RMSPE(a)	22.84	1.03	0.53	1091.3%	109.3%	32.5%	1233.1%
	MAPE(a)	22.76	1.08	0.56	1087.3%	119.2%	38.5%	1245.0%
	U(a)	23.59	1.10	0.40	1130.6%	122.7%	0.0%	1253.3%
	RMSE(a)	23.86	1.15	0.40	1144.5%	132.8%	0.0%	1277.3%
	MAE(a)	26.38	1.17	0.48	1275.8%	137.4%	19.0%	1432.2%

Table C2

Comparison of calibration error for the IDM and for the FVDM with SIGMOID spacing policy, calibrated against AstaZero trajectories.

	IDM (AstaZero)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	2.67	0.51	0.39	2.6%	6.3%	483.9%	492.8%
	U(s)	2.61	0.49	0.40	0.4%	3.4%	491.4%	495.3%
	U(s,v)	2.62	0.49	0.40	0.6%	3.4%	491.3%	495.4%
	NRMSE(s,v,a)	2.60	0.49	0.40	0.0%	3.3%	492.2%	495.5%
	RMSE(s)	2.60	0.49	0.40	0.0%	3.3%	492.3%	495.6%
	NRMSE(s,v)	2.60	0.49	0.40	0.0%	3.1%	492.5%	495.6%
	RMSPE_tilde(s)	2.62	0.50	0.40	0.7%	4.0%	491.3%	496.1%
	RMSPE_tilde(s,v)	2.62	0.49	0.40	0.6%	3.7%	491.9%	496.3%
	MAE(s)	2.67	0.49	0.40	2.4%	3.6%	492.0%	498.0%
	RMSPE(s)	2.70	0.50	0.40	3.5%	4.9%	490.0%	498.5%
	RMSPE(s,v)	2.70	0.50	0.40	3.5%	4.7%	490.4%	498.6%
	MAPE(s,v)	2.69	0.50	0.40	3.1%	4.2%	491.7%	499.0%
	MAPE(s)	2.72	0.50	0.40	4.5%	4.5%	490.5%	499.5%
	RMSPE(s,stdV)	2.74	0.50	0.40	5.3%	4.3%	492.1%	501.7%
	RMSPE_tilde(v)	4.19	0.48	0.40	61.0%	0.1%	496.4%	557.6%
	RMSE(v)	4.22	0.48	0.40	62.3%	0.0%	495.7%	558.0%
	U(v)	4.23	0.48	0.40	62.5%	0.0%	495.8%	558.3%
	MAE(v)	4.25	0.48	0.40	63.1%	0.6%	495.8%	559.5%
	MAPE(v)	4.46	0.48	0.40	71.3%	1.1%	495.7%	568.1%
	RMSPE(v)	4.64	0.49	0.40	78.2%	3.8%	491.9%	573.9%
	RMSPE(s,v,a)	28.77	1.58	0.31	1005.0%	231.6%	354.4%	1591.0%
	MAPE(s,v,a)	57.25	2.64	0.28	2099.2%	454.7%	320.0%	2873.8%
	RMSE(stdV)	80.23	1.66	0.33	2981.7%	248.6%	393.1%	3623.4%
	RMSPE(stdV)	80.23	1.66	0.33	2981.7%	248.6%	393.1%	3623.4%
	MAPE(a)	730.29	5.51	0.09	27951.1%	1056.4%	40.7%	29048.2%
	RMSPE(a)	935.38	5.73	0.09	35828.8%	1103.6%	30.2%	36962.6%
	U(a)	1053.95	5.45	0.08	40383.4%	1045.4%	13.4%	41442.2%
	RMSE(a)	1470.26	6.13	0.07	56374.2%	1188.0%	0.0%	57562.2%
	MAE(a)	1650.12	6.28	0.07	63282.7%	1219.1%	2.2%	64504.0%

	SIGMOID (AstaZero)	CALIBRATION ERRORS			RELATIVE CALIBRATION ERRORS			
		$\overline{RMSE}(s)$ [m]	$\overline{RMSE}(v)$ [m/s]	$\overline{RMSE}(a)$ [m/s ²]	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	NRMSE(s,v,a)	2.54	0.49	0.42	0.0%	7.5%	31.1%	38.6%
	U(s,v)	2.57	0.48	0.42	1.2%	7.0%	32.6%	40.7%
	RMSPE_tilde(s,v)	2.57	0.49	0.43	1.4%	7.6%	32.8%	41.8%
	RMSE(s)	2.54	0.49	0.43	0.0%	8.0%	34.0%	42.0%
	MAE(s)	2.58	0.48	0.43	1.7%	7.5%	33.6%	42.8%
	NRMSE(s,v)	2.58	0.49	0.43	1.6%	8.8%	34.5%	44.9%
	U(s)	2.57	0.50	0.43	1.5%	9.7%	35.3%	46.5%
	MAPE(s)	2.63	0.49	0.43	3.9%	8.5%	35.0%	47.4%
	MAPE(s,v)	2.64	0.49	0.43	4.3%	8.3%	35.1%	47.7%
	RMSPE(s,v)	2.71	0.50	0.43	7.0%	11.5%	32.9%	51.4%
	RMSPE_tilde(s)	2.65	0.51	0.44	4.4%	12.5%	36.0%	53.0%
	RMSPE(s)	2.73	0.51	0.43	7.6%	12.3%	33.4%	53.3%
	RMSPE(s,stdV)	2.71	0.53	0.45	6.9%	17.8%	41.2%	65.8%
	U(s,v,a)	2.99	0.58	0.39	17.8%	29.0%	21.1%	67.9%
	RMSPE(v)	4.59	0.49	0.42	80.9%	8.1%	30.5%	119.6%
	RMSPE_tilde(v)	5.05	0.45	0.43	99.1%	0.7%	32.8%	132.5%
	RMSPE(s,v,a)	4.22	0.77	0.36	66.6%	71.3%	13.5%	151.4%
	MAPE(v)	5.62	0.45	0.43	121.7%	0.0%	33.9%	155.5%
	MAPE(s,v,a)	4.33	0.80	0.37	70.9%	77.6%	14.5%	163.0%
	U(v)	6.23	0.45	0.43	145.8%	0.1%	33.9%	179.8%
	RMSE(v)	6.62	0.45	0.43	161.1%	0.0%	33.8%	194.8%
	MAE(v)	7.85	0.45	0.43	209.5%	0.8%	35.7%	246.0%
	RMSE(stdV)	24.77	0.79	0.47	876.8%	74.5%	46.5%	997.7%
	RMSPE(stdV)	27.08	0.85	0.45	968.1%	88.9%	41.5%	1098.5%
	U(a)	33.78	0.87	0.32	1232.2%	92.3%	0.1%	1324.6%
	RMSE(a)	33.77	0.88	0.32	1231.8%	94.2%	0.0%	1326.0%
	MAE(a)	33.80	0.88	0.32	1232.9%	95.4%	0.2%	1328.6%
	RMSPE(a)	34.55	0.89	0.32	1262.8%	97.5%	0.4%	1360.7%
	MAPE(a)	35.23	0.90	0.32	1289.6%	99.7%	1.2%	1390.5%

Table C3

Relative calibration errors for the four models, calibrated against NGSIM I80-1 trajectories.

IDM (NGSIM)		RELATIVE CALIBRATION ERRORS			
		$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	8.1%	7.9%	142.8%	158.8%
	NRMSE(s,v,a)	2.5%	6.6%	151.3%	160.4%
	U(s)	0.4%	10.3%	160.7%	171.4%
	U(s,v)	1.5%	6.8%	163.1%	171.4%
	NRMSE(s,v)	1.4%	6.8%	165.0%	173.2%
	RMSE(s)	0.0%	11.3%	168.0%	179.3%
	MAE(s)	2.7%	12.7%	164.4%	179.7%
	RMSPE_tilde(s,v)	6.2%	11.0%	166.3%	183.5%
	RMSPE_tilde(s)	3.8%	14.0%	177.4%	195.1%
	RMSPE_tilde(v)	53.4%	3.2%	144.5%	201.1%
	MAE(v)	57.8%	2.0%	142.9%	202.7%
	RMSE(v)	41.3%	0.0%	162.0%	203.3%
	U(v)	40.9%	0.1%	166.2%	207.2%
	MAPE(v)	65.3%	10.1%	131.9%	207.3%
	MAPE(s,v)	10.9%	9.4%	188.2%	208.5%
	RMSPE(v)	70.0%	11.3%	137.7%	219.0%
	MAPE(s)	14.4%	13.1%	193.4%	220.9%
	RMSPE(s,stdV)	21.6%	15.5%	201.8%	238.9%
	RMSPE(s,v)	16.4%	13.5%	212.0%	241.9%
	RMSPE(s)	17.1%	15.6%	217.3%	250.0%
	RMSPE(stdV)	154.2%	35.5%	214.8%	404.5%
	RMSPE(stdV)	165.5%	41.3%	231.7%	438.5%
	RMSPE(s,v,a)	291.4%	126.8%	97.2%	515.5%
	MAPE(s,v,a)	319.2%	137.9%	76.0%	533.1%
	U(a)	1183.1%	269.0%	3.6%	1455.7%
	RMSE(a)	1531.2%	326.0%	0.0%	1857.1%
	MAPE(a)	1464.3%	281.7%	117.0%	1863.0%
	RMSPE(a)	1513.4%	283.5%	137.6%	1934.5%
	MAE(a)	1763.4%	320.8%	31.9%	2116.1%
GIPPS (NGSIM)		RELATIVE CALIBRATION ERRORS			
		$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	10.4%	10.3%	75.2%	95.9%
	NRMSE(s,v,a)	1.5%	8.9%	88.1%	98.5%
	NRMSE(s,v)	0.0%	6.7%	97.9%	104.6%
	U(s,v)	1.7%	6.9%	97.7%	106.2%
	U(s)	0.0%	10.9%	101.5%	112.2%
	RMSPE_tilde(s,v)	2.8%	9.6%	101.0%	113.4%
	RMSE(s)	0.0%	12.1%	101.6%	113.7%
	MAE(s)	2.4%	10.3%	105.5%	118.2%
	RMSPE_tilde(s)	4.9%	14.8%	106.5%	126.3%
	MAPE(s,v)	14.0%	15.2%	102.8%	132.0%
	MAPE(s)	13.8%	14.2%	106.0%	134.0%
	U(v)	41.4%	0.3%	92.7%	134.5%
	RMSE(v)	44.6%	0.0%	91.7%	136.3%
	RMSPE_tilde(v)	47.6%	3.0%	98.4%	149.0%
	RMSPE(s,v)	20.5%	20.3%	110.6%	151.4%
	MAE(v)	68.6%	2.1%	94.8%	165.6%
	RMSPE(s,stdV)	30.0%	19.5%	117.2%	166.8%
	RMSPE(s)	21.5%	21.7%	127.9%	171.1%
	MAPE(v)	75.6%	6.5%	100.7%	182.8%
	RMSPE(v)	89.4%	14.0%	109.0%	212.4%
	RMSPE(stdV)	187.2%	36.8%	95.6%	319.6%
	RMSPE(stdV)	170.2%	48.0%	108.8%	327.0%
	RMSPE(s,v,a)	274.7%	108.7%	28.0%	411.4%
	MAPE(s,v,a)	302.3%	118.1%	27.5%	447.9%
	U(a)	740.3%	158.3%	2.1%	900.7%
	RMSE(a)	903.5%	180.7%	0.0%	1084.2%
	RMSPE(a)	930.4%	185.7%	38.0%	1154.1%
	MAE(a)	1898.7%	288.9%	37.2%	2224.8%
	MAPE(a)	1794.0%	283.5%	190.1%	2267.6%
CTH (NGSIM)		RELATIVE CALIBRATION ERRORS			
		$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	9.7%	5.6%	67.0%	82.4%
	NRMSE(s,v,a)	4.5%	5.3%	81.5%	91.2%
	U(s,v)	4.1%	6.7%	104.4%	115.2%
	RMSE(s)	0.0%	10.5%	110.6%	121.0%
	NRMSE(s,v)	5.2%	7.1%	120.0%	132.3%
	RMSPE_tilde(s,v)	5.5%	6.7%	123.5%	135.7%
	RMSE(v)	63.6%	0.0%	91.3%	154.9%
	MAE(v)	62.3%	2.6%	92.0%	156.9%
	U(v)	67.4%	0.8%	102.5%	170.7%
	RMSPE(s,v)	20.7%	11.9%	144.3%	177.0%
	RMSPE_tilde(v)	59.4%	0.3%	117.5%	177.2%
	U(s)	8.2%	20.0%	165.3%	193.4%
	RMSE(v)	60.8%	6.9%	126.8%	194.5%
	MAPE(v)	69.2%	7.6%	125.4%	202.2%
	MAE(s)	17.2%	29.5%	156.2%	202.9%
	RMSPE_tilde(s)	10.9%	23.4%	179.8%	214.1%
	MAPE(s,v)	24.7%	23.5%	168.8%	217.0%
	RMSPE(s)	17.2%	23.4%	187.6%	228.2%
	MAPE(s)	26.8%	34.2%	212.0%	273.0%
	RMSPE(s,v,a)	233.6%	85.7%	28.1%	347.4%
	RMSPE(s,stdV)	45.0%	46.6%	334.1%	425.7%
	U(a)	328.5%	105.9%	2.2%	436.5%
	MAPE(s,v,a)	238.4%	101.0%	10.4%	441.8%
	RMSPE(a)	333.5%	110.4%	12.2%	456.1%
	RMSE(a)	341.4%	118.6%	0.0%	460.0%
	RMSPE(stdV)	143.0%	49.1%	269.2%	461.3%
	MAE(a)	351.8%	112.8%	24.2%	488.7%
	MAPE(a)	351.2%	113.9%	93.3%	558.4%
	RMSE(stdV)	152.5%	57.4%	351.0%	561.0%
SIGMOID (NGSIM)		RELATIVE CALIBRATION ERRORS			
		$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	9.2%	8.3%	85.5%	102.9%
	NRMSE(s,v,a)	0.4%	7.1%	97.1%	104.6%
	NRMSE(s,v)	2.9%	12.2%	142.0%	157.0%
	U(s,v)	2.0%	13.0%	154.5%	169.5%
	RMSE(v)	62.9%	0.0%	106.9%	169.9%
	U(s)	0.8%	17.4%	158.4%	176.5%
	RMSPE(s,v)	18.1%	15.0%	148.9%	182.0%
	MAE(s)	8.3%	21.9%	161.7%	191.9%
	RMSPE_tilde(s,v)	10.1%	13.7%	168.6%	192.3%
	MAE(v)	79.6%	3.9%	113.6%	197.0%
	RMSE(s)	0.0%	17.9%	180.1%	198.0%
	RMSPE_tilde(v)	78.4%	2.6%	124.4%	205.4%
	RMSPE_tilde(s)	6.5%	17.8%	181.6%	205.9%
	MAPE(v)	75.6%	6.3%	134.6%	216.4%
	U(v)	82.3%	2.9%	141.3%	226.4%
	RMSPE(v)	73.1%	11.3%	144.6%	229.0%
	MAPE(s,v)	27.8%	27.9%	195.0%	250.7%
	MAPE(s)	24.2%	30.1%	210.9%	265.2%
	RMSPE(s)	23.1%	30.5%	221.8%	275.4%
	RMSPE(s,v,a)	251.4%	74.8%	46.2%	372.4%
	MAPE(s,v,a)	196.6%	92.3%	176.9%	465.8%
	RMSPE(s,stdV)	49.1%	54.6%	370.4%	474.1%
	U(a)	431.4%	125.7%	1.5%	558.6%
	RMSE(stdV)	168.7%	72.1%	345.9%	586.7%
	RMSE(a)	454.9%	142.3%	0.0%	597.2%
	RMSPE(a)	474.6%	144.5%	35.0%	654.2%
	MAE(a)	506.3%	146.3%	16.1%	668.8%
	MAPE(a)	440.9%	115.1%	140.5%	696.5%
	RMSPE(stdV)	224.5%	87.9%	392.0%	704.4%

Table C4

Relative calibration errors for the four models, calibrated against Hefei trajectories.

	IDM (Hefei)	RELATIVE CALIBRATION ERRORS					GIPPS (Hefei)	RELATIVE CALIBRATION ERRORS			
		$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$			$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
CALIBRATION GOFs	U(s,v,a)	3.9%	4.9%	496.9%	505.7%	CALIBRATION GOFs	U(s,v,a)	7.6%	8.1%	179.2%	195.0%
	NRMSE(s,v,a)	0.0%	2.5%	521.5%	524.0%		NRMSE(s,v)	1.6%	5.1%	190.4%	197.2%
	RMSPE_tilde(s,v)	4.7%	4.2%	533.0%	541.9%		NRMSE(s,v,a)	3.0%	6.2%	189.2%	198.3%
	U(v)	26.7%	0.1%	518.7%	545.5%		U(s,v)	5.0%	7.3%	187.3%	199.6%
	MAPE(s,v)	10.2%	5.0%	530.6%	545.8%		RMSE(s)	0.0%	6.3%	193.6%	199.9%
	MAE(v)	32.7%	0.2%	514.6%	547.5%		U(s)	3.7%	7.9%	188.4%	200.0%
	RMSE(v)	34.3%	0.0%	514.3%	548.6%		MAE(s)	3.2%	7.9%	192.6%	203.8%
	MAPE(s)	14.4%	8.5%	529.9%	552.8%		RMSPE_tilde(s,v)	7.8%	9.2%	193.1%	210.1%
	RMSPE(s,v)	16.2%	6.1%	531.0%	553.3%		RMSPE_tilde(s)	11.4%	14.3%	190.6%	216.3%
	MAPE(v)	40.2%	0.9%	514.9%	556.0%		MAPE(s)	17.3%	11.4%	197.7%	226.4%
	NRMSE(s,v)	1.2%	5.1%	549.8%	556.0%		MAPE(s,v)	21.5%	12.9%	197.5%	231.8%
	RMSPE(v)	44.5%	2.5%	519.4%	566.4%		RMSPE(s)	26.3%	16.3%	197.9%	240.5%
	RMSPE(s)	25.9%	11.8%	531.8%	569.5%		RMSE(v)	55.6%	0.0%	184.9%	240.5%
	RMSPE_tilde(v)	52.6%	1.3%	522.4%	576.2%		U(v)	55.5%	0.2%	184.9%	240.6%
	RMSPE_tilde(s)	7.6%	8.5%	562.1%	578.2%		RMSPE(s,v)	29.9%	16.2%	198.4%	244.5%
	RMSE(s)	0.0%	5.1%	575.2%	580.4%		MAPE(v)	58.4%	2.0%	186.2%	246.5%
	U(s,v)	1.7%	4.1%	575.2%	581.0%		MAE(v)	63.3%	1.2%	186.5%	251.0%
	RMSPE(s,stdV)	22.3%	13.6%	550.2%	586.0%		RMSPE_tilde(v)	69.5%	1.9%	187.6%	259.0%
	MAE(s)	2.4%	5.8%	580.6%	588.8%		RMSPE(v)	68.7%	8.1%	189.9%	266.7%
	U(s)	2.8%	7.8%	579.6%	590.3%		RMSPE(s,stdV)	44.0%	26.7%	208.6%	279.3%
CALIBRATION GOFs	RMSPE(stdV)	176.9%	57.1%	654.5%	888.5%		RMSPE(stdV)	173.2%	53.2%	215.3%	441.6%
	RMSPE(s,v,a)	291.2%	146.4%	474.2%	911.8%		RMSPE(s,v,a)	249.1%	116.6%	167.8%	533.5%
	RMSPE(stdV)	236.4%	78.3%	621.3%	936.0%		RMSE(stdV)	338.1%	95.6%	215.5%	649.2%
	MAPE(s,v,a)	350.0%	175.4%	434.9%	960.2%		MAPE(s,v,a)	328.8%	148.3%	174.5%	651.6%
	U(a)	6269.9%	551.8%	32.6%	6854.2%		U(a)	2356.0%	293.9%	5.5%	2655.4%
	MAPE(a)	7835.8%	582.3%	103.4%	8521.5%		RMSPE(a)	2606.1%	315.9%	21.9%	2943.9%
	RMSPE(a)	8575.6%	621.7%	44.6%	9241.9%		MAE(a)	2649.9%	315.1%	7.4%	2972.4%
	MAE(a)	12554.6%	699.1%	1.5%	13255.2%		MAPE(a)	2634.4%	319.0%	21.8%	2975.2%
	RMSE(a)	12639.1%	699.1%	0.0%	13338.2%		RMSE(a)	2828.0%	345.6%	0.0%	3173.6%
CALIBRATION GOFs	CTH (Hefei)	RELATIVE CALIBRATION ERRORS				CALIBRATION GOFs	SIGMOID (Hefei)	RELATIVE CALIBRATION ERRORS			
	U(s,v,a)	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$		U(s,v,a)	$\overline{RE}(s)$	$\overline{RE}(v)$	$\overline{RE}(a)$	$\sum_i \overline{RE}(i)$
	NRMSE(s,v,a)	3.9%	12.4%	147.1%	163.4%		U(s,v)	2.4%	6.9%	210.7%	220.0%
	NRMSE(s,v)	0.7%	10.3%	156.4%	167.4%		NRMSE(s,v,a)	0.0%	6.3%	223.4%	229.6%
	RMSE(s)	2.6%	8.5%	177.0%	188.2%		NRMSE(s,v)	1.5%	7.0%	228.5%	237.0%
	U(s,v)	0.0%	14.4%	181.1%	195.5%		U(s)	1.8%	8.6%	245.6%	255.9%
	U(s,v)	6.7%	10.1%	178.9%	195.7%		NRMSE(s,v)	3.3%	9.2%	256.8%	269.3%
	U(s)	6.3%	13.8%	176.6%	196.7%		MAE(s)	4.9%	9.5%	259.6%	274.1%
	RMSPE_tilde(s,v)	18.3%	14.2%	189.3%	221.7%		RMSE(s)	0.0%	13.0%	261.3%	274.2%
	MAE(s)	15.0%	18.2%	196.2%	229.4%		RMSPE_tilde(s,v)	16.4%	10.2%	255.2%	281.8%
	RMSPE_tilde(s)	21.2%	17.9%	190.7%	229.8%		MAPE(s,v)	34.8%	10.8%	250.2%	295.8%
	U(v)	54.9%	2.0%	175.1%	232.0%		RMSPE_tilde(s)	18.8%	14.2%	268.5%	301.5%
	RMSPE(v)	63.9%	4.6%	172.3%	240.8%		MAPE(v)	78.5%	0.9%	243.0%	322.3%
	RMSPE_tilde(v)	65.7%	2.0%	174.9%	242.7%		RMSPE(s)	48.4%	19.0%	261.1%	328.5%
	RMSPE(s,v)	40.4%	18.9%	184.9%	244.2%		RMSPE(v)	101.2%	1.6%	244.1%	346.9%
	RMSE(v)	74.3%	0.0%	178.1%	252.4%		RMSE(v)	117.4%	0.0%	239.6%	357.0%
	RMSPE(s)	46.4%	22.3%	184.1%	252.8%		RMSPE_tilde(v)	107.7%	5.9%	244.1%	357.8%
	MAPE(v)	72.7%	5.8%	181.3%	259.8%		U(v)	107.9%	5.7%	258.9%	372.6%
	MAE(v)	78.6%	2.4%	183.4%	264.4%		RMSPE(s,stdV)	46.6%	23.2%	307.0%	376.7%
	MAPE(s)	49.6%	26.2%	196.6%	272.5%		MAE(v)	131.9%	6.2%	245.5%	383.6%
	RMSPE(s,stdV)	44.7%	23.8%	217.0%	285.5%		MAPE(s)	55.5%	21.5%	306.7%	383.7%
CALIBRATION GOFs	MAPE(s,v)	57.2%	28.6%	258.7%	344.5%		RMSPE(s,v)	60.5%	22.3%	339.5%	422.3%
	RMSPE(s,v,a)	98.4%	68.3%	334.0%	500.7%		RMSPE(s,v,a)	111.9%	65.4%	461.5%	638.7%
	MAPE(s,v,a)	103.0%	75.1%	335.1%	513.2%		MAPE(s,v,a)	112.1%	68.6%	488.3%	669.0%
	RMSPE(stdV)	118.1%	61.3%	442.5%	621.9%		RMSPE(stdV)	159.6%	70.3%	640.1%	870.0%
	RMSE(stdV)	144.4%	69.3%	503.8%	717.5%		MAPE(a)	414.6%	99.3%	404.7%	918.7%
	MAPE(a)	580.6%	121.4%	274.4%	976.3%		RMSE(stdV)	166.5%	68.3%	685.0%	919.9%
	RMSPE(a)	839.3%	170.6%	212.5%	1222.4%		MAE(a)	1423.6%	135.9%	217.1%	1776.5%
	MAE(a)	1283.7%	165.2%	139.9%	1588.8%		RMSPE(a)	1372.3%	168.1%	244.0%	1784.5%
	RMSE(a)	1657.2%	200.8%	0.0%	1857.9%		RMSE(a)	2457.0%	260.4%	0.0%	2717.4%
	U(a)	2005.9%	272.9%	11.5%	2290.3%		U(a)	2850.0%	337.1%	7.8%	3194.9%

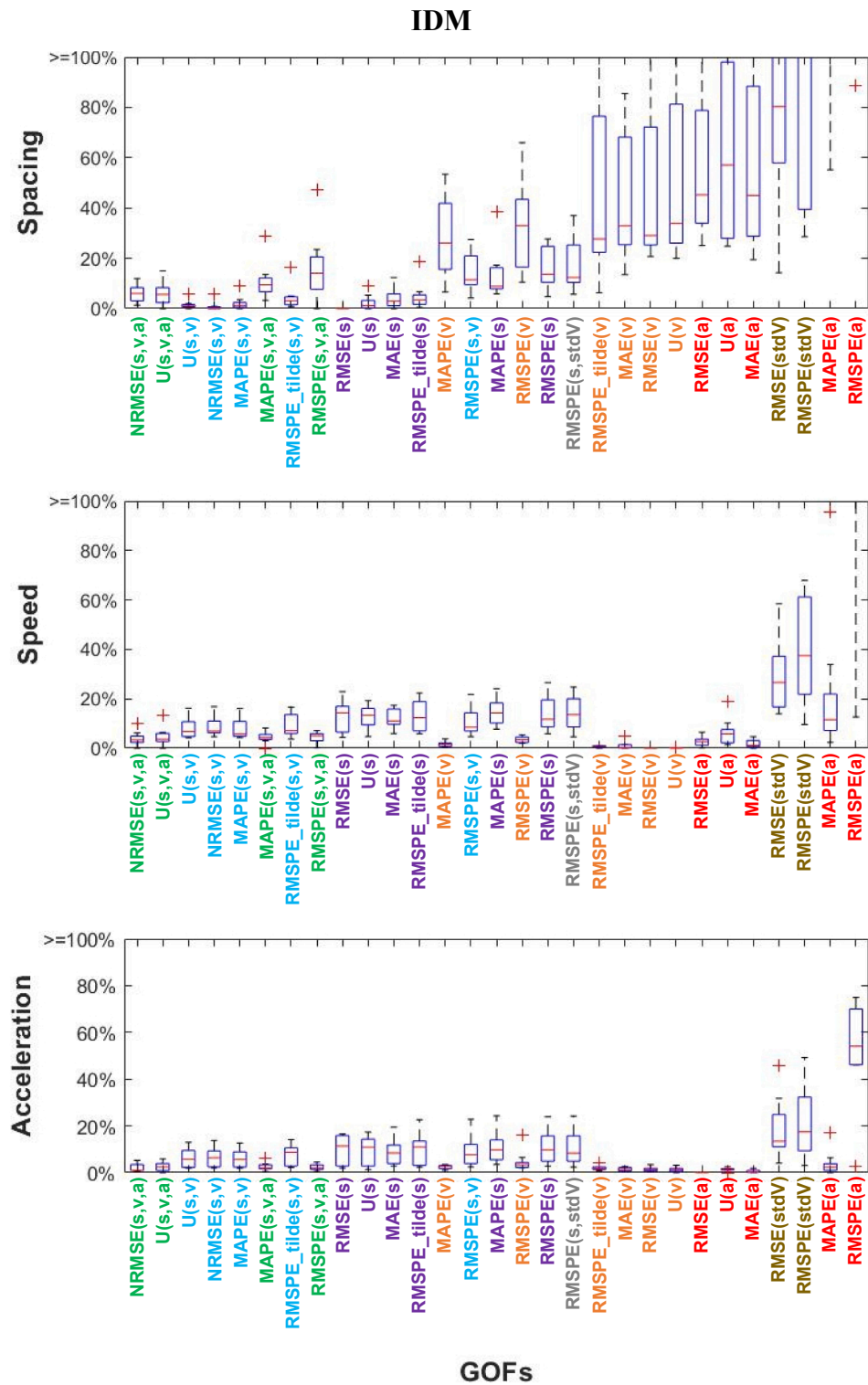


Fig. C1. IDM relative calibration error box plots for Naples trajectories. The $GoF(MoP)$ functions are sorted according to Table C1. Colours identify $GoFs$ with the same MoP or combination of $MoPs$.

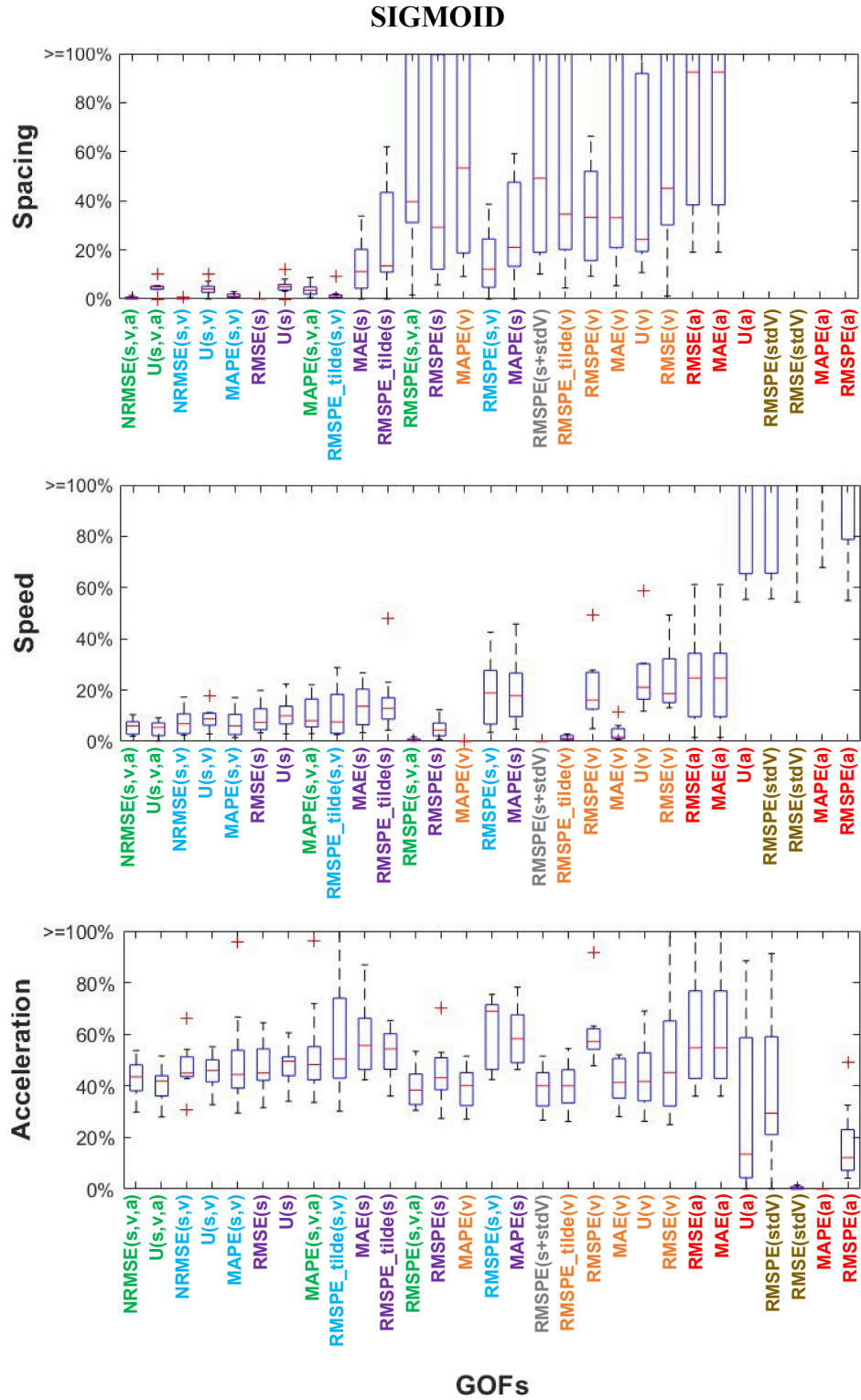


Fig. C2. FVDM with SIGMOID spacing policy relative calibration error box plots for Naples trajectories. The *GoF(MoP)* functions are sorted according to Table C1. Colours identify *GoFs* with the same *MoP* or combination of *MoPs*.

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Appendix A

Here follows the list of the target journals in the literature review (we regard these journals as the premier venues for researchers in the traffic flow community, thus representing the state-of-the-art):

- Transportation Research series (e.g., Transportation Research Part B, Part C, etc.)
- Transportation Science
- Transportation Research Record
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Control System Technology
- IEEE Transactions on Intelligent Vehicles
- IEEE journal on Automatic Control
- Vehicle system dynamics
- IET Intelligent Transport Systems
- Transportation Planning and Technology
- Transportmetrica A, and B
- Physica A
- Physical Review E
- Accident Analysis and Prevention
- Nonlinear Dynamics
- European Transport Research Review
- Philosophical Transactions of the Royal Society A
- Journal of Intelligent Transportation Systems
- Journal of Traffic and Transportation Engineering

Appendix B

In the following, there are some notes about the classification of the selected papers in the *GoF/MoP* notation in Table 2.

Hoogendoorn and Hoogendoorn (2010a, 2010b), Hoogendoorn et al. (2010), Kim et al. (2013), Rhoades et al. (2016), Xu and Laval (2020): The authors applied the *MLE* method. Under the assumption of Gaussian errors, the *MLE* method is equivalent to the *GLS* method.

Zhu et al. (2018a, 2018b, 2020), Chen et al. (2020), Kurtc and Treiber (2020): The authors applied a normalized *SSE*($\cdot$) over the square of the observed measure. From an optimization viewpoint, such *GoF* function is identical to the *NRMSE*($\cdot$), being one of its linear transformations.

Chong et al. (2011), Rahman et al., 2015, Li et al. (2016), Xu et al. (2019): The authors applied $SSE(\cdot) = \sum_{t=1}^T \left[(\cdot)^{sim}(t) - (\cdot)^{obs}(t) \right]^2$. From an optimization viewpoint, such *GoF* function is identical to *RMSE*($\cdot$), being one of its linear transformations.

van Hinsbergen et al. (2015), Huang et al. (2018a), Li et al. (2018, 2020), Wang et al. (2018), Yu et al. (2018), Zhou et al. (2017): The authors applied *MSE*($\cdot$). From an optimization viewpoint, such *GoF* function is identical to *RMSE*($\cdot$), being one of its linear transformations.

Yu and Shi (2014): The authors applied *MSPE*($\cdot$). From an optimization viewpoint, such *GoF* function is identical to *RMSPE*($\cdot$), being one of its linear transformations.

Sangster et al. (2013), Zhang et al. (2019): The authors applied the root operator to the squared terms, rather than to the mean squared terms. From an optimization viewpoint, such *GoF* function is identical to the Theil's Inequality Coefficient *U*, being one of its linear transformations.

Milanes and Shladover (2014): The authors applied $IAE(\cdot) = \sum_{t=1}^T \left| (\cdot)^{sim}(t) - (\cdot)^{obs}(t) \right|$. Such *GoF* function is identical to the *MAE*($\cdot$), being one of its linear transformations.

Appendix C

See Tables C1–C4.

See Figs. C1 and C2.

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